Repetitionless

The offline documentation can be hard to follow so using the online documentation is recommended which you can find at https://docs.wilschack.dev/repetitionless. You can still use this if you need though, it has all the same information

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1. Introduction

1.1 Features Video

https://youtu.be/BqwM6js7TkU

1.2 Description

Repetitionless is a set of shaders that removes texture tiling using multiple different toggleable techniques including:

- Voronoi noise and cell edge sampling to split up the texture and smooth between cells
- Random scaling and rotation between cells
- Triplanar sampling
- Variation texture support to add variation to the material
- Distance blending to either change the tiling and offset or material entirely at a set distance range
- Material Blending to overlay a separate material ontop of the main one by different noise functions or a custom mask

1.3 Contents

2. Installation

2.1 Download

Requirements:

- Unity 2021.3+

Repetitionless can be downloaded from:

- Unity Asset Store
- Itch.io

2.2 Installation

2.2.1 Unity Asset Store

Install it the same you would any other unity package:

1. Open the Unity Package Manager
2. Select Repetitionless under My Assets
3. Click Download, then Import after it is finished

2.2.2 Itch.io

You can install the package by either:

- Dragging the .unitypackage file into the project window
- Importing it through `Assets > Import Package > Custom Package`

3. Getting Started With Repetitionless

3.1 Tutorial Video

https://youtu.be/Ujk6ZyCmWak

3.2 Welcome Screen

When first importing the asset you will be greeted with a welcome screen with buttons to:

- View Documentation
- View and Reporting issues
- Import samples

To open this window again, you can find it in the toolbar at `Window > Repetitionless > Open Window`

3.3 Sample Scenes

The asset comes with 3 example scenes that you can view to see the different features of repetitionless and how to implement the materials

For instructions on accessing the samples:

View the Samples page

3.4 Using A Repetitionless Material

There are details you need to know when using the materials:

View the Material Details Page

For instructions on converting regular lit materials to repetitionless materials:

View the Converting Materials Page

For instructions on how to use the regular repetitionless shader:

View the Using Regular Materials Page

For instructions on how to use the layered repetitionless shader or a terrain:

View the Using Layered Materials Page

To view what each material property does:

View the Material Properties Page

3.5 Painting

If you want details on how to paint layers onto objects:

View the Painting Page

3.6 Integrations

If you want to use any of the asset integrations, details are in their respective pages:

For instructions on using Repetitionless terrain materials on a MapMagic terrain:

View the MapMagic Page

3.7 Creating Shaders

You can view the Sub Graphs and Shader API documentation in the contents on the left of the page

If you want to create a shader graph or code shader using the features from the repetitionless materials:

View the Shader Creation page

3.8 Creating Scripts

You can view the Sub Graphs and Shader API documentation in the contents on the left of the page

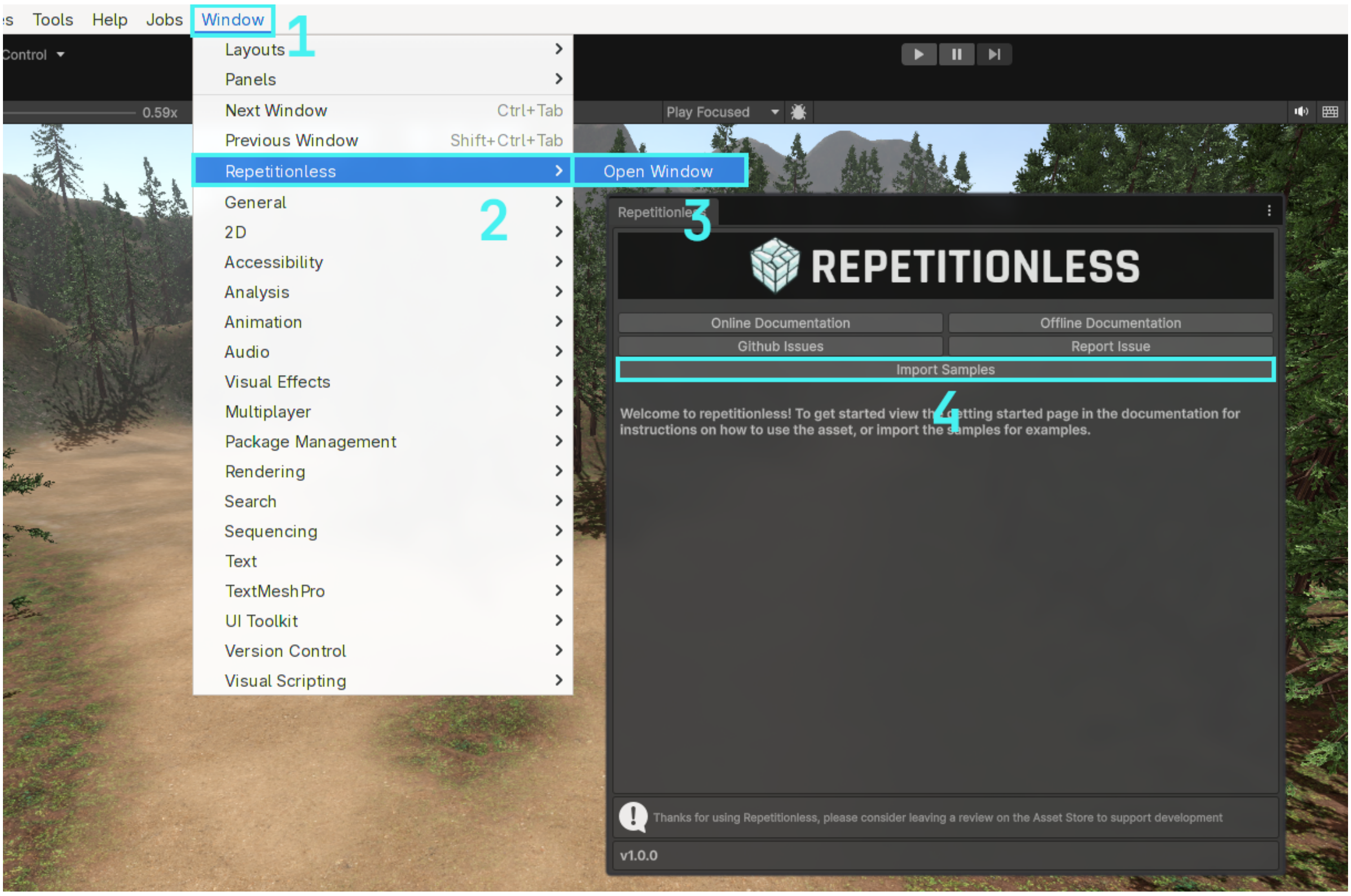
If you want to create scripts or inspectors using the features from the repetitionless materials inspectors:

View the Scripting page

4. Samples

Samples are located in a folder which is hidden by default. To import the samples:

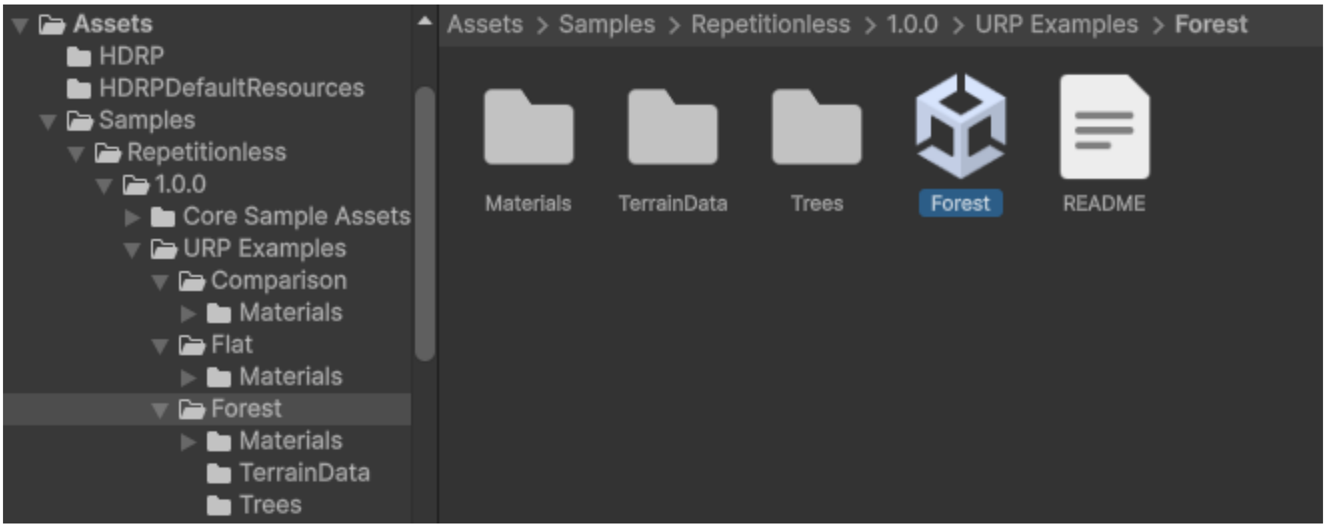
1. Open the windows tab in the toolbar
2. Navigate to `Repetitionless`
3. Click `Open Window`
4. Click the import samples button and it will automatically import your current render pipelines samples



4.1 Opening The Samples

Samples will now be located in `Assets/Samples/Repetitionless/<CurrentVersion>/<RP> Examples`

To open the sample scenes you can go to their respective folders and open the scene

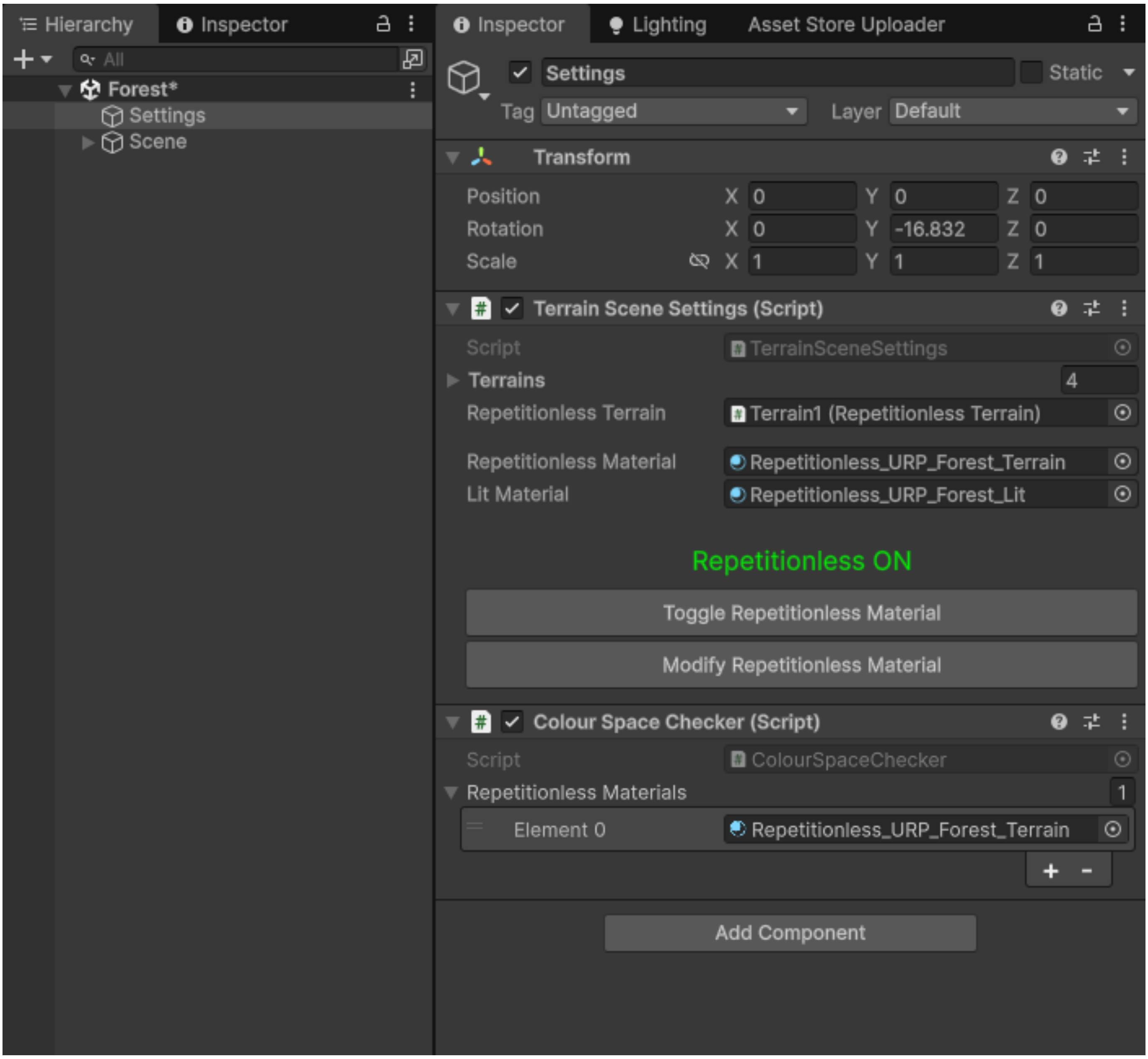


The Core Sample Assets folder stores the base assets for all the samples, this must be imported for the samples to work

4.2 Sample Settings

The samples have a settings object that has buttons to:

- Toggle between the repetitionless and regular lit materials
- Modify and view the material settings used



(The Colour Space Checker is used to make sure the textures are packed in the correct colour space, it is not needed after the scene is opened for the first time)

5. Material Details

5.1 Data Folder

All the properties and textures are stored in a folder along side the material named
`<Material Name>_RepetitionlessData`

This folder is automatically handled and you do not need to worry about it

This folder will automatically be moved and deleted with the material but will not be copied so if you need to copy a material you will need to manually copy the folder aswell

5.2 Texture Packing

All the textures are packed into texture arrays for less samples in the shader which are stored in the data folder along side the material. These can be modified by clicking the **Array Settings button** in the Material Properties section

Arrays are split into:

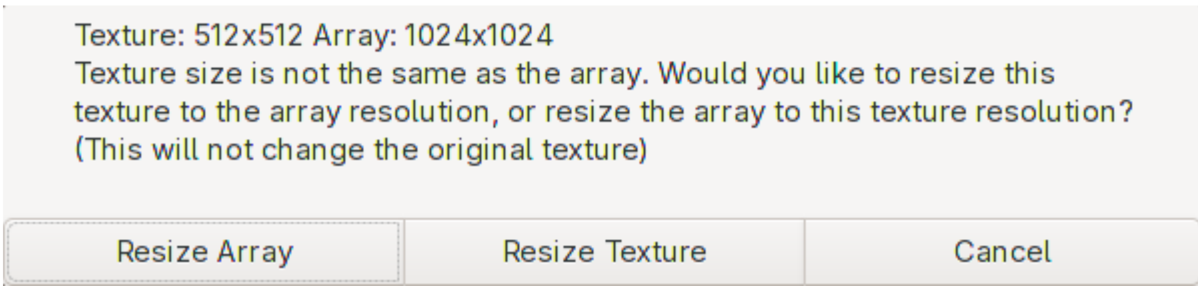
- AVTextures: Albedo (rgb), Variation (a)
- NSOTextures: Normal (rg), Smoothness (b), Occlussion (a)
- EMTextures: Emission (rgb), Metallic (a)
- BMTextures: Blend Mask (r)

Only the required arrays are created, and only the used textures are stored

The downside to texture arrays is that they can only have one resolution each array of textures, but this is automatically handled. This is also per array so for example the albedo textures can be a different resolution to the normal textures

When you assign a texture that is not the same resolution as the array, there is a popup that lets you either:

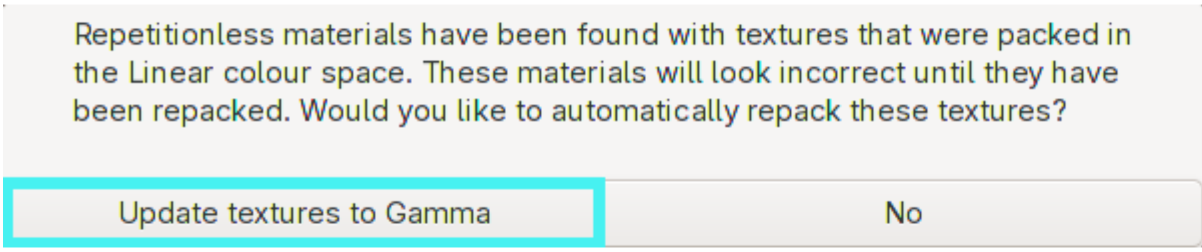
- Resize all the textures in the array to the resolution of the new texture
- Resize the texture to the same resolution as the array



5.3 Colour Space

sRGB textures (Albedo/Colour) are packed with the set colour space in mind. They will need to be repacked when changing the colour space

There is a popup when changing the colour space if any textures need to be repacked that will automatically repack the textures for you.
Make sure to select **Update Textures to \<Colour Space>** for the textures to look correct



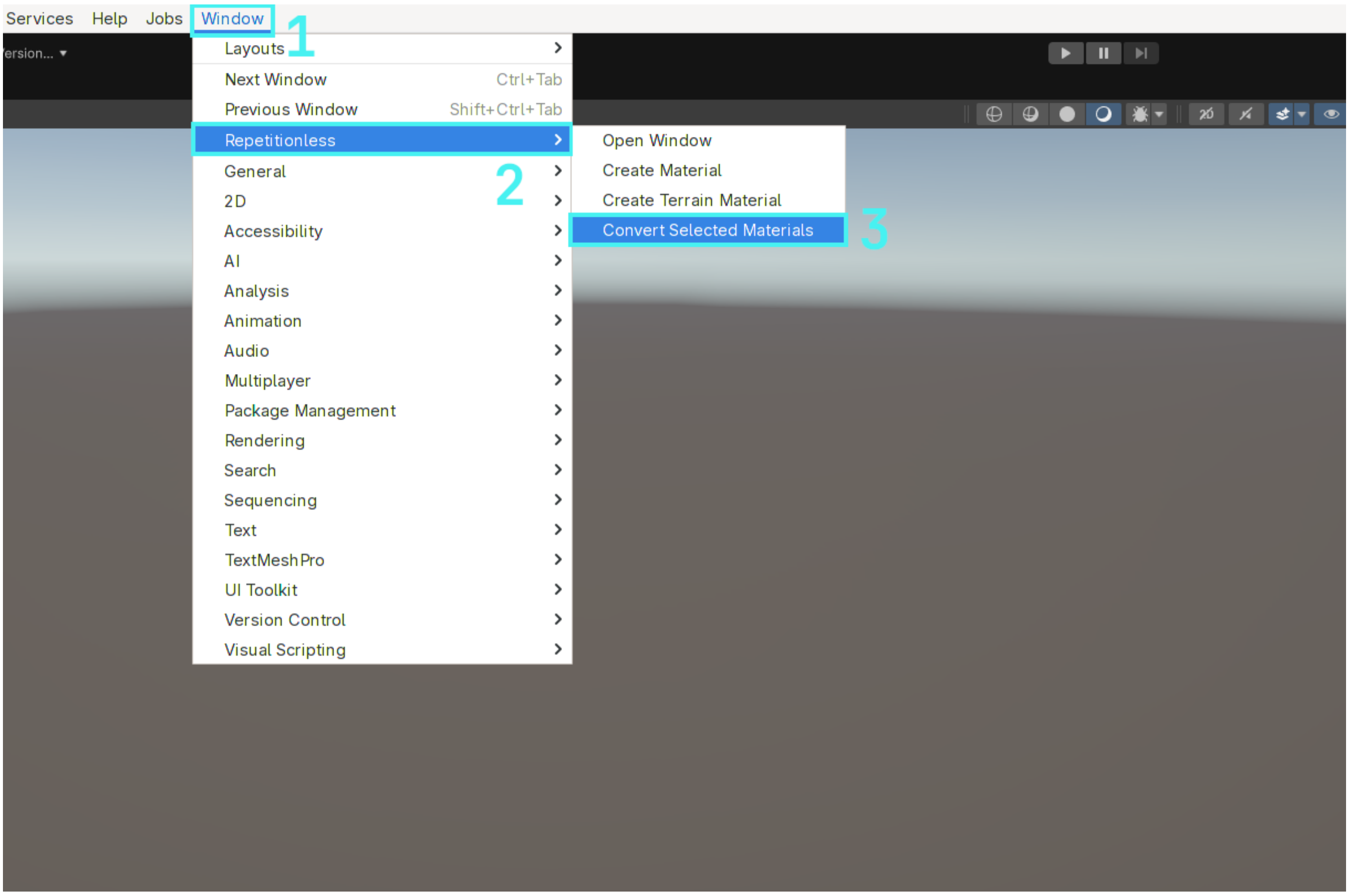
To view what each property does, visit the Material Properties page

6. Converting Materials

6.1 Converting A Material

To convert a regular lit material to a repetitionless material:

1. Select the materials you want to convert
2. Open the `Window` tab in the toolbar, or the `Create` menu in the project window
3. Navigate to `Repetitionless`
4. Click `Convert Selected Materials`
5. The materials will be created next to the original materials with the suffix "_repetitionless"



Important Details:

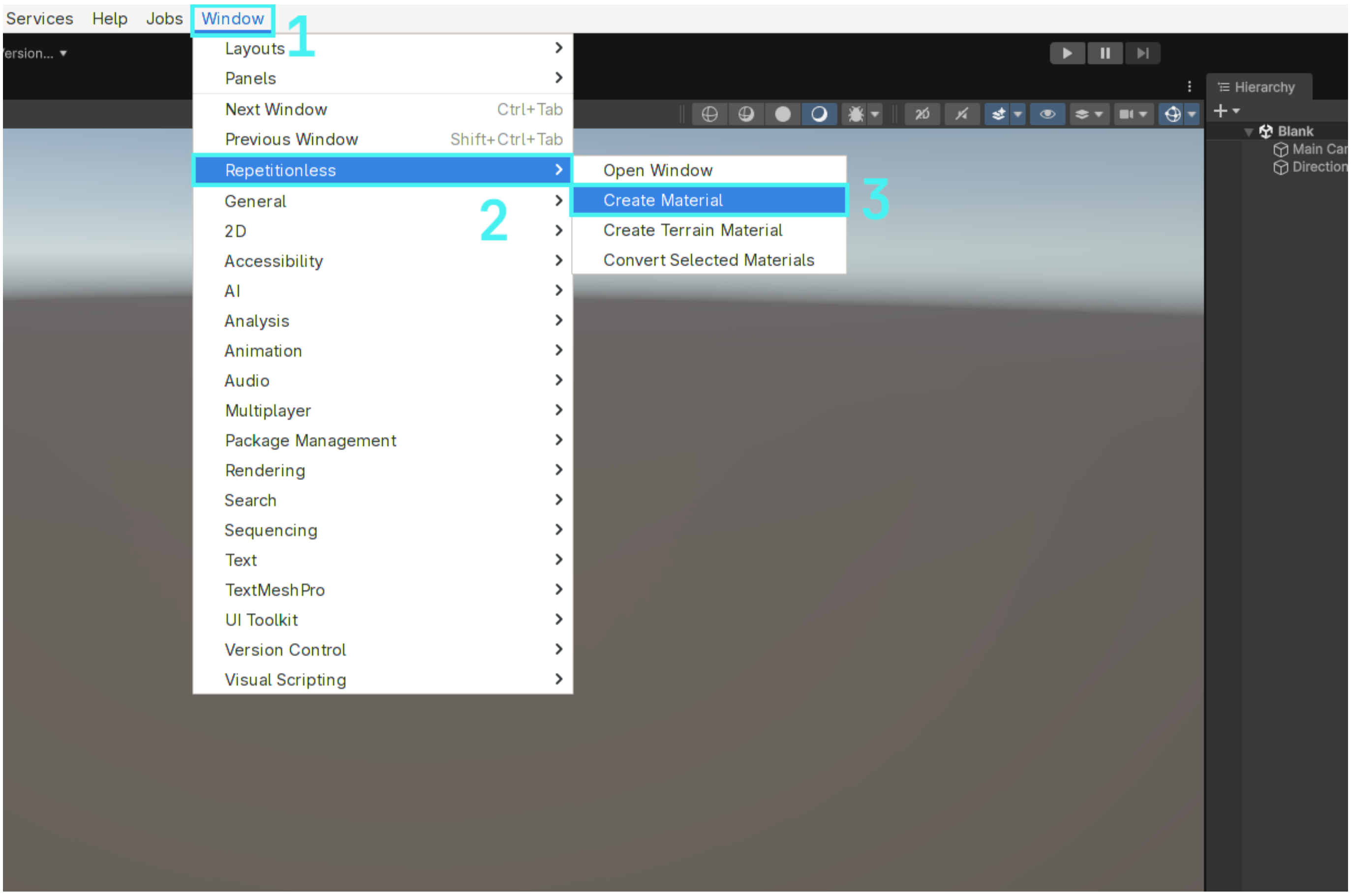
- This will always convert to the same render pipeline as the input material
- This works for BIRP, URP, & HDRP standard lit materials, any others will not work

7. Using Regular Materials

7.1 Creating A Material

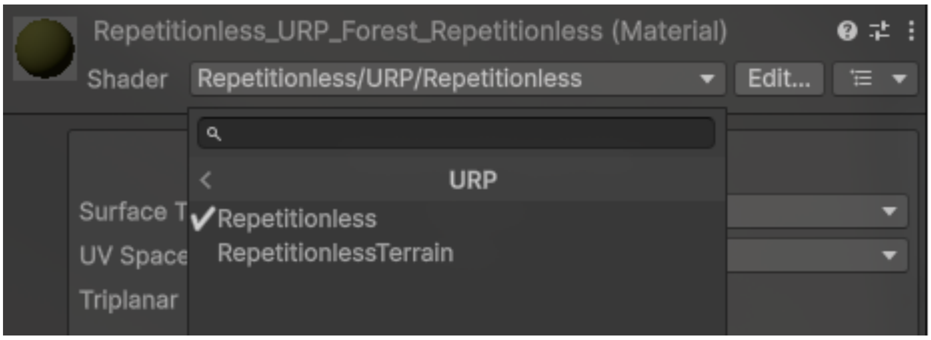
7.1.1 Automatic

1. Open the `Window` tab in the toolbar, or the `Create` menu in the project window
2. Navigate to `Repetitionless`
3. Click `Create Material`
4. The material will be created in the current folder in the project window



7.1.2 Manual

1. Create a material
2. Select the shader dropdown
3. Navigate to `Repetitionless`
4. Select your render pipeline
5. Select `Repetitionless`

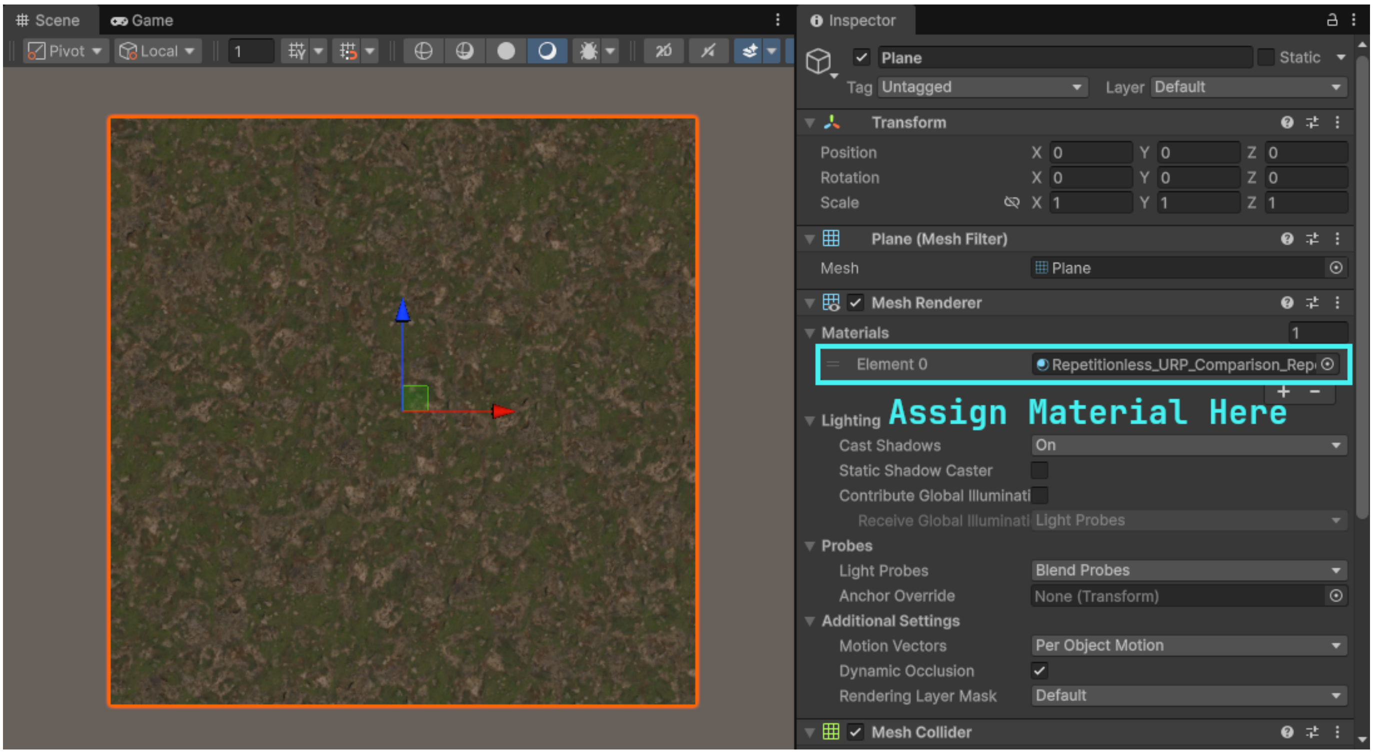


Important Details:

- Render pipelines can be switched without losing data. ex. Changing a Repetitionless material from BIRP to its URP version will keep all the settings
- Material data is stored in a folder along side the material named "{MaterialName}_Data". This is automatically moved and deleted with the material but it will need to be manually copied when copying the material

7.2 Using The Material

You can just assign repetitionless material to any Mesh Renderer and it will be applied to that object



7.3 Configuration

All the materials can be configured through its inspector by selecting the material and viewing the inspector window

To view what each property does, visit the Material Properties page

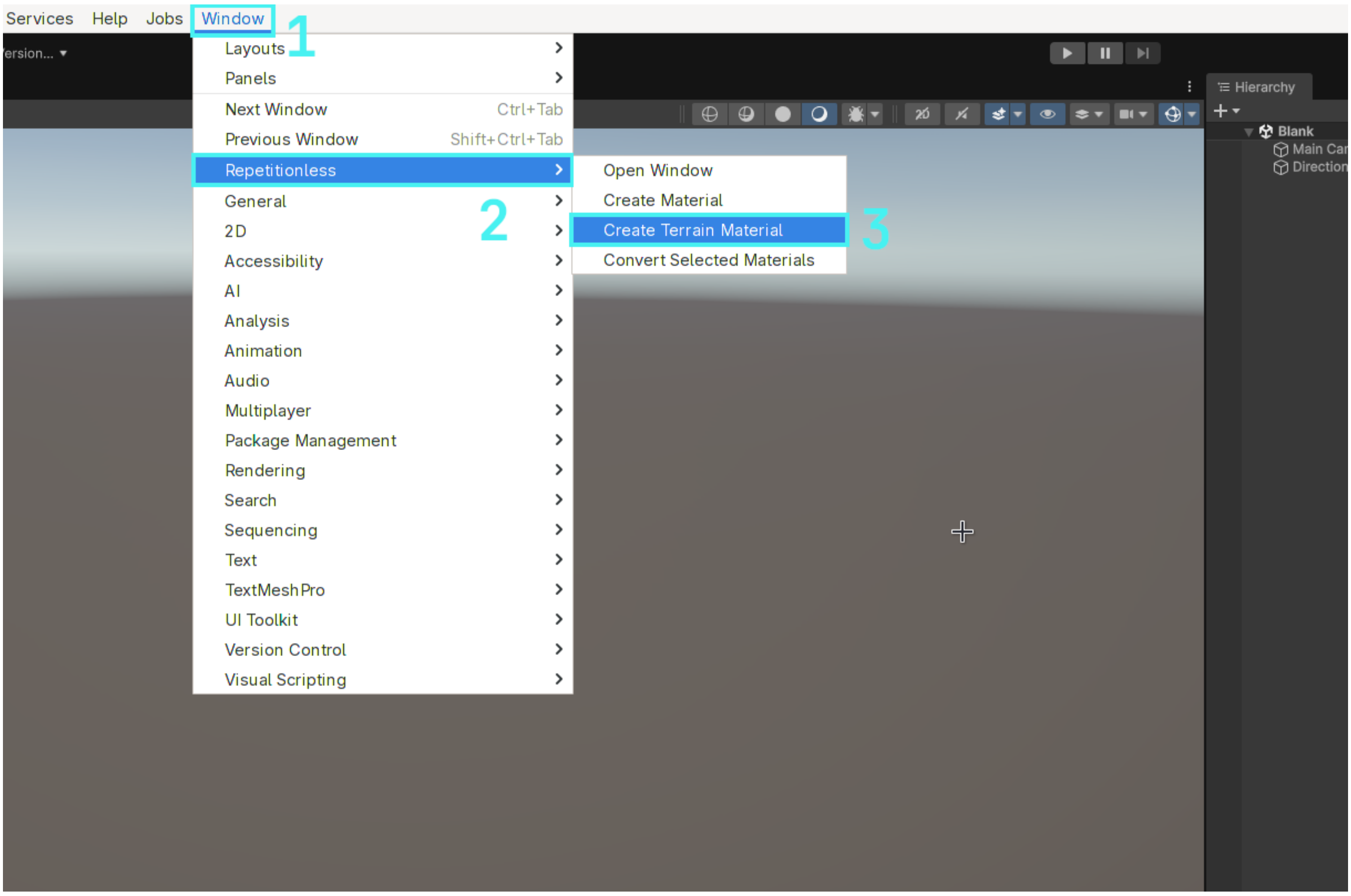
8. Using Layered Materials

The layered material is used to display up to 32 different material layers on one material. Layer positions can either be updated with control textures or synced with a terrain. Instructions are below for both

8.1 Creating A Layered Material

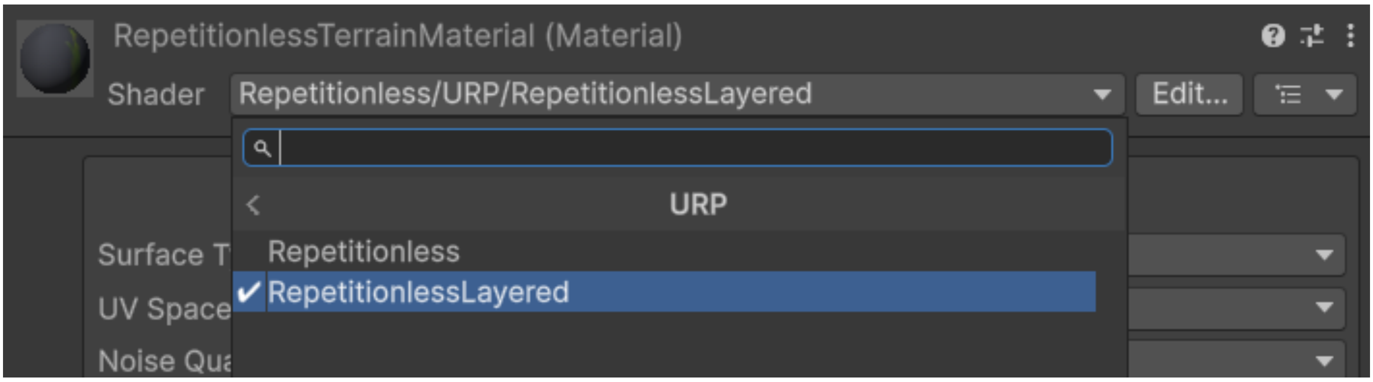
8.1.1 Automatic

1. Open the `Window` tab in the toolbar, or the `Create` menu in the project window
2. Navigate to `Repetitionless`
3. Click `Create Layered Material`
4. The material will be created in the current folder in the project window



8.1.2 Manual

1. Create a material
2. Select the shader dropdown
3. Navigate to `Repetitionless`
4. Select your render pipeline
5. Select `RepetitionlessLayeredTerrain` or `RepetitionlessLayeredLit`



Important Details:

- Render pipelines can be switched without losing data. ex. Changing a Repetitionless material from BIRP to its URP version will keep all the settings
- Material data is stored in a folder along side the material named "{MaterialName}_RepetitionlessData". This is automatically moved and deleted with the material but it will need to be manually copied when copying the material
- The `LayeredTerrain` shader is used when the mode is set to Terrain Layers, `LayeredLit` is used when the mode is set to anything else. This can be changed in the inspector, so when creating the material you can select either.

8.2 Using Terrain Layers

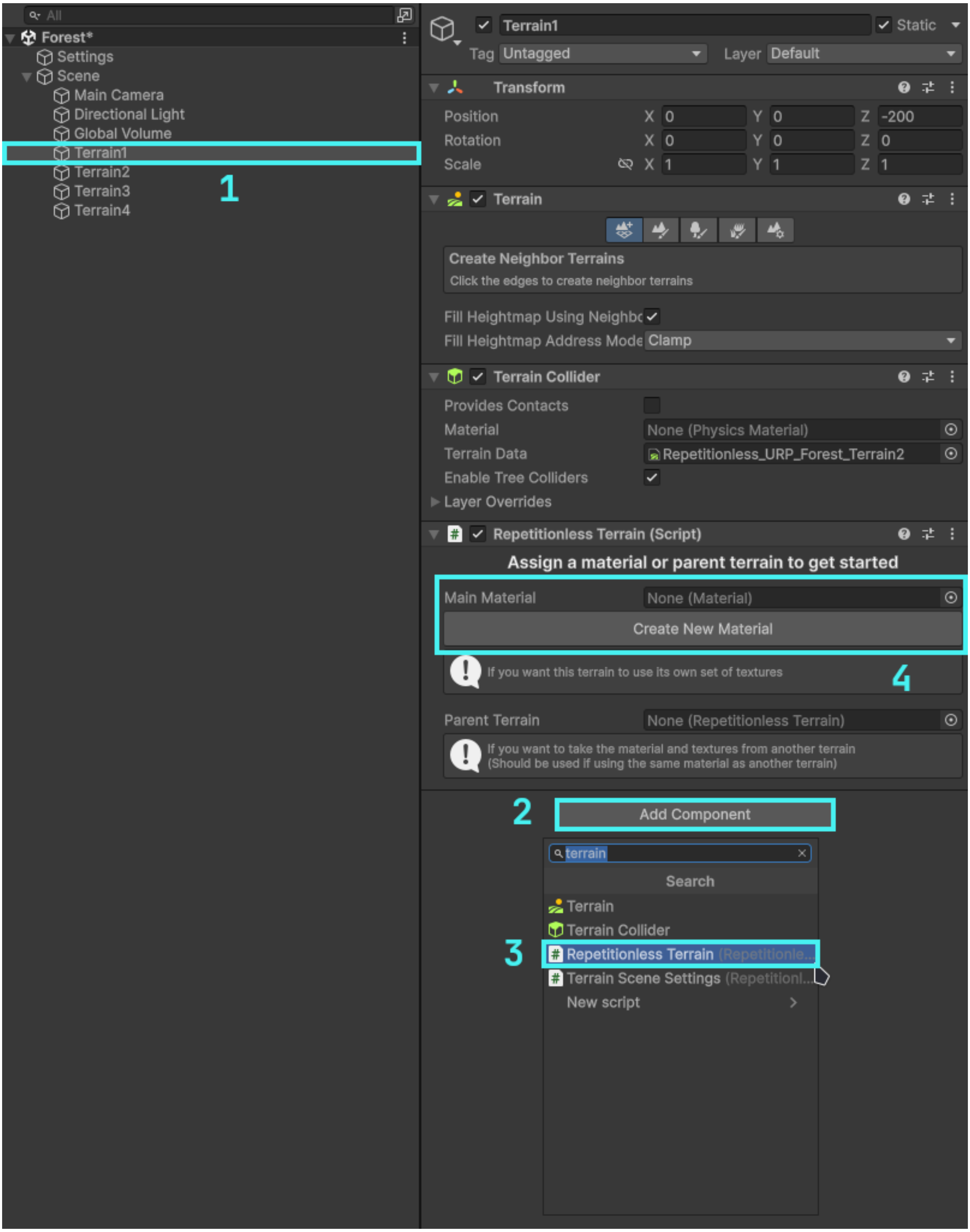
To use a terrain with the repetitionless material you have to add a `RepetitionlessTerrain` component to the terrain which can be done as follows:

1. Select the terrain you want to use
2. Select `Add Component`
3. Select `Repetitionless Terrain`

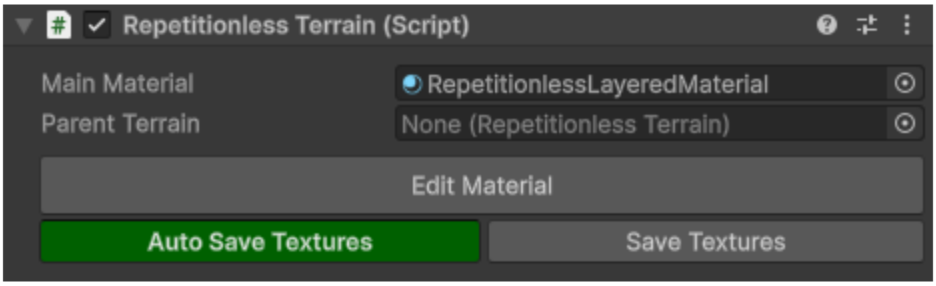
To add a material you can either:

- Click the `Create New Material` button
- Assign a material in the `Main Material` field

Note the the material field will only allow Repetitionless Layered materials. Assigning a material will also automatically set its layer mode to Terrain Layers


All
Forest*
Settings
Scene
Main Camera
Directional Light
Global Volume
Terrain1
Terrain2
Terrain3
Terrain4
1
Terrain1
Static
Tag Untagged
Layer Default
Transform
Position X 0 Y 0 Z -200
Rotation X 0 Y 0 Z 0
Scale X 1 Y 1 Z 1
Terrain
Create Neighbor Terrains
Click the edges to create neighbor terrains
Fill Heightmap Using Neighbc
Fill Heightmap Address Mode Clamp
Terrain Collider
Provides Contacts
Material None (Physics Material)
Terrain Data Repetitionless_URP_Forest_Terrain2
Enable Tree Colliders
Layer Overrides
Repetitionless Terrain (Script)
Assign a material or parent terrain to get started
Main Material None (Material)
Create New Material
If you want this terrain to use its own set of textures
4
Parent Terrain None (Repetitionless Terrain)
If you want to take the material and textures from another terrain
(Should be used if using the same material as another terrain)
2
Add Component
terrain
Search
Terrain
Terrain Collider
3
Repetitionless Terrain (Repetitionle...
Terrain Scene Settings (Repetitionl...
New script

8.2.1 Using The Terrain Script



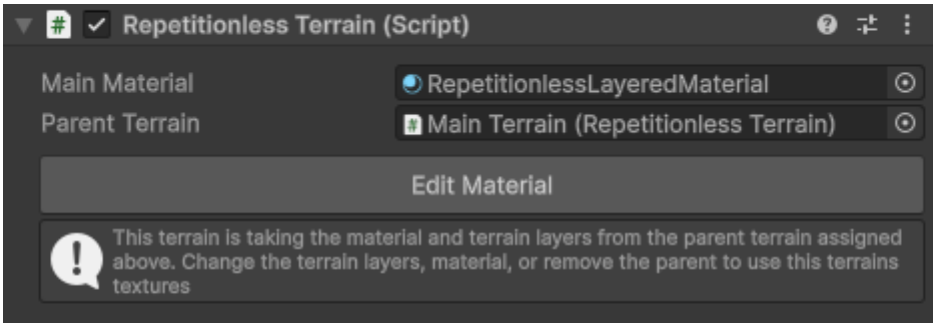
Setting	Description
Main Material	The material applied to the terrains
Parent Terrain	The terrain to take the material and terrain layers from. See below for details
Edit Material	Opens the inspector for the selected material
Auto Save Textures	Toggles if the material layers are automatically updated if the terrain layers change
Save Textures	Reapplies the terrain materials if required, and syncs the terrain layers to the material. Click this if you have any issues with the terrain

That is basically all you need to do! The script will handle the material assigning and texture syncing for you

With Auto Save Textures enabled, the terrain will sync whenever you update the terrain layers, updating the materials textures. With it disabled, it will not sync the textures until you click the Save Textures button.

If you ever have any errors with the terrain, first see if clicking the Save Textures button fixes your issue. If it doesnt feel free to submit a bug report

8.2.2 Using Multiple Terrains

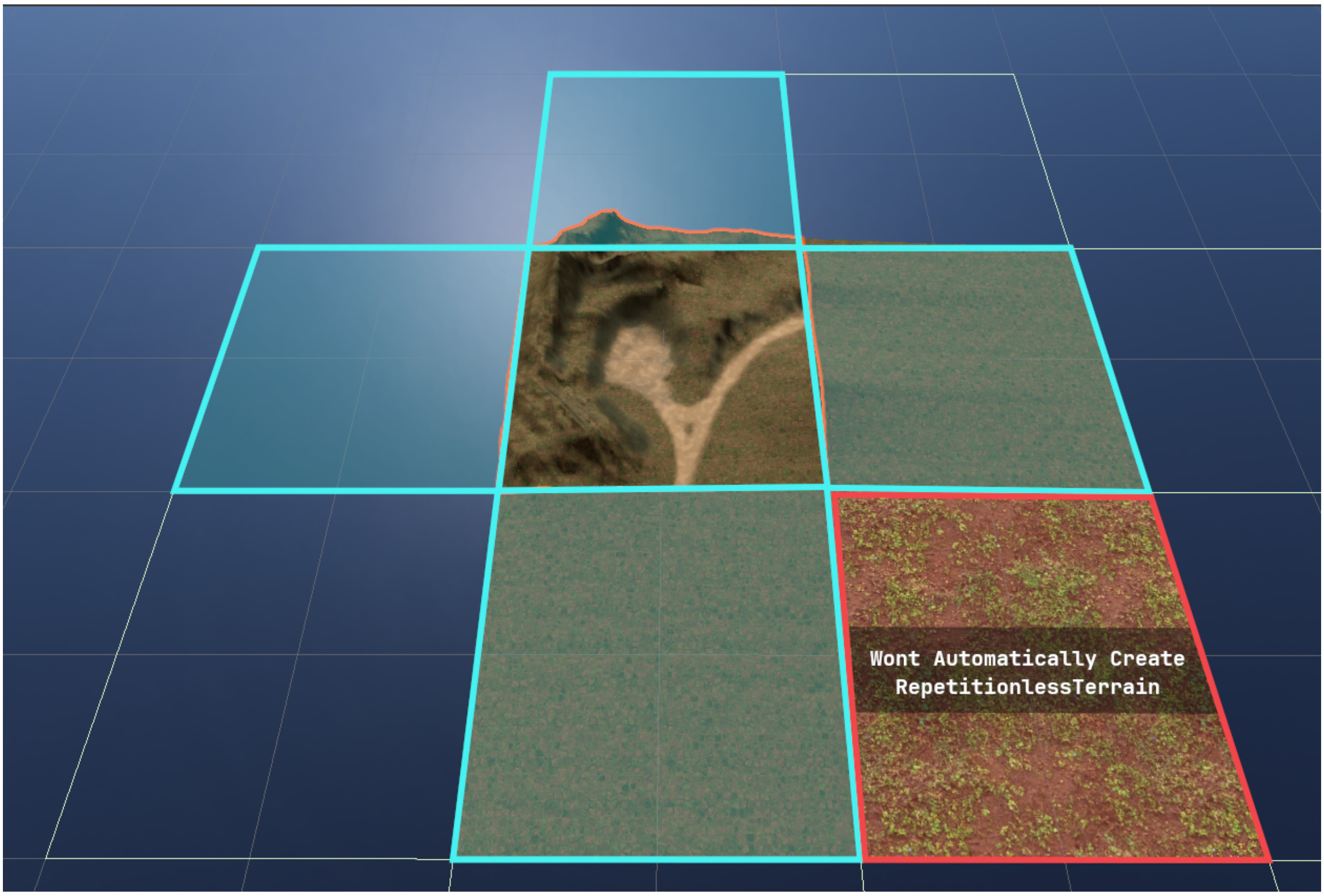


The script also has a Parent Terrain field that when set will update that terrain to use the same material and layers from the parent. Keep in mind that when setting a parent terrain it will overwrite the current terrain layers you have set.

If a terrain has a parent, it will automatically update when the parent terrain updates, so using the save textures button on the main terrain will update all the children aswell.

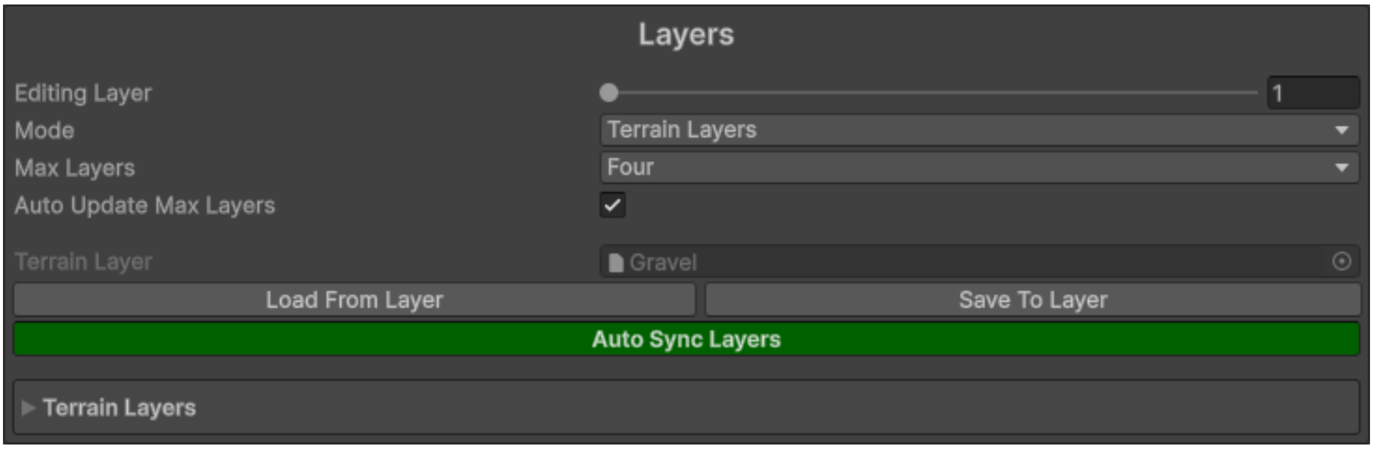
Important Details:

- One parent terrain can be assigned to as many child terrains as you like
- Terrain Materials should only be synced to one set of terrain layers, but you can use the parent terrain field to get around that
- The parent terrain will automatically detach when you either change the terrain layers, or update the material



On terrains with the RepetitionlessTerrain, when using the Create Neighbour Terrains tool and creating a directly adjacent terrain, it will automatically add a RepetitionlessTerrain component to that terrain with the selected one as the parent. This will not work for non-adjacent terrains though as shown in the image above

8.2.3 Editing The Terrain Layers



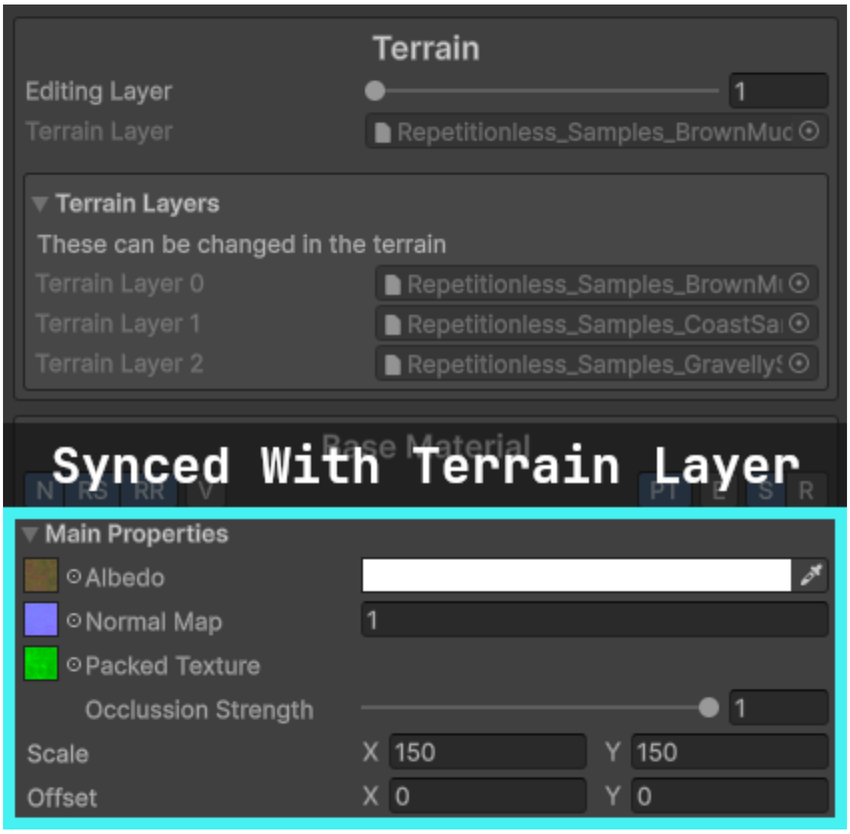
Setting	Description
Editing Layer	The layer currently being edited in the inspector
Mode	The mode of this material. This changes the shader depending on the option with Terrain Layers changing it to a terrain compatible shader
Max Layers	The max layers allowed on this material. Set lower when possible for performance improvements
Auto Update Max Layers	Automatically updates the above Max Layers to fit your assigned layer count

After modifying any field in the terrain layer and saving, it will automatically update those properties in any material that is linked to that layer if the auto sync is enabled. You can manually load or save to and from the layer with the buttons

The properties effected and what they link to include:

- Diffuse Map > Base Albedo Texture
- Normal Map > Base Normal Map
- Normal Scale > Base Normal Scale
- Mask Map > Base Packed Texture (Will enable packed texture when assigned)
- Metallic > Base Metallic
- Smoothness > Base Smoothness
- Tiling > Base Tiling
- Offset > Base Offset

8.2.4 Editing The Material



If a terrain has been assigned, when editing the material it will show the terrain layers that it is linked to and each layer can be edited individually.

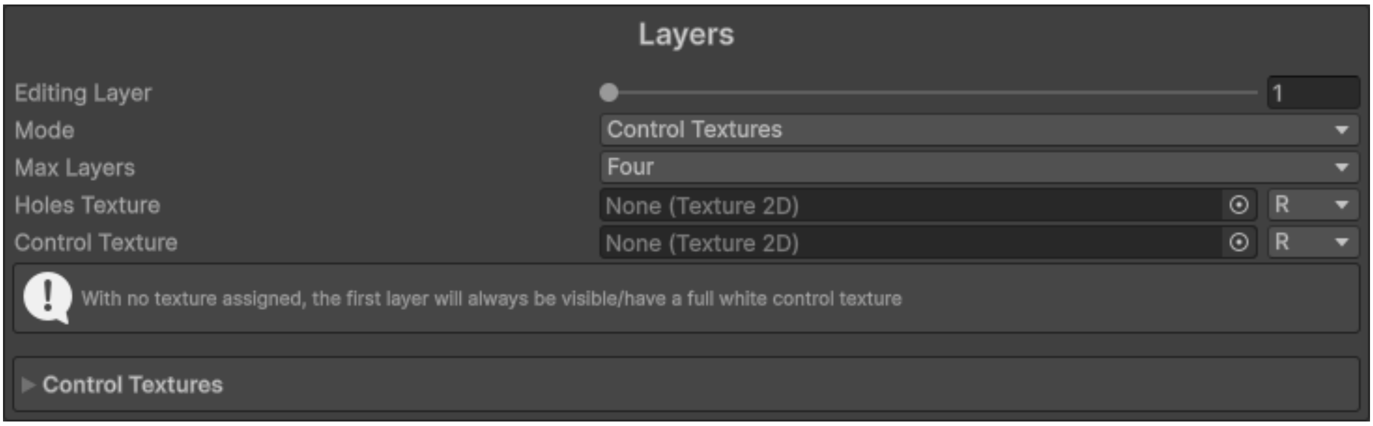
Updating the Main Properties in the Base Material that have an equivalent in the terrain layer (listed above) will also automatically update the terrain layer

Layers in the repetitionless material correspond to the order of the assigned terrain layers as shown in the image above

The shader supports up to 32 terrain layers. Any more wont work and will appear white

8.3 Using Control Textures

To use control textures, first make sure the layer mode is set to `Control Textures`



Setting	Description
Editing Layer	The layer currently being edited in the inspector
Mode	The mode of this material. This changes the shader depending on the option with Terrain Layers changing it to a terrain compatible shader
Max Layers	The max layers allowed on this material. Set lower when possible for performance improvements
Holes Texture	The holes texture that will be used for the material. It will read from the selected channel
Control Texture	The control texture that will be used for this layer. It will read from the selected channel

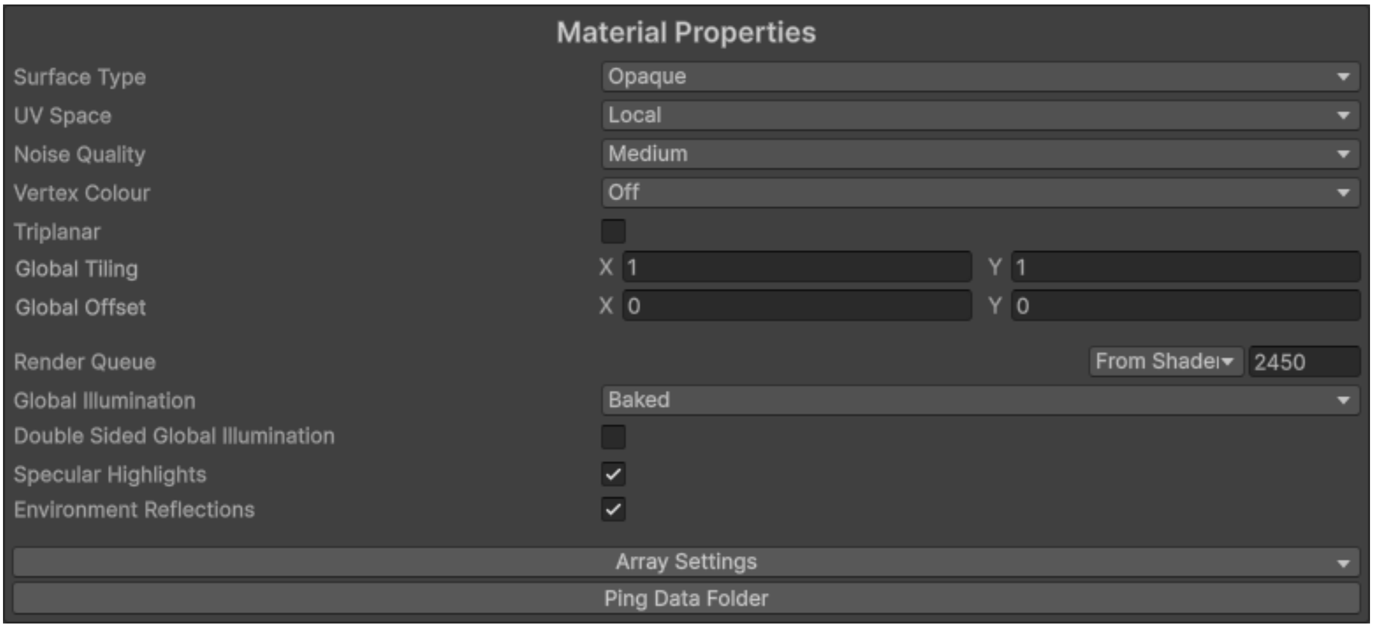
The control textures for a specific pixel must add up to 1. For example, if you want a spot to be the second layer, the second control texture will have a value of 1 in that spot where other layers will have a value of 0.

Example of two control textures where the left is layer 1, and the right layer 2

Everything else is the same as the regular repetitionless material. To view what each property does, visit the Material Properties page

9. Material Properties

9.1 Material Properties



Has all the general properties for the material

Property	Description
Surface Type	The surface type of the material 0: Opaque 1: Cutout 2: Transparent
UV Space	Sets the UV Space 0: Local 1: World *Note that with world space UVs you will require larger Tiling*
Noise Quality	How the noise is sampled High: Dynamically samples noise *(Required for non-repeating noise and the angle offset setting)* Medium: Uses the pre-rendered 4k texture Low: Uses the pre-rendered 1k texture
Vertex Colour	How the vertex colour is blended with the output colour 0: Off 1: Multiply 2: Additive 3: Subtractive 4: Overwrite
Triplanar	Enables triplanar sampling and forces world UVs *Note that the first time this is set for a material it will take some time to finish as it will recompile the shader, it will be instant for subsequent toggles*
Global Tiling	This is multiplied with each materials tiling
Global Offset	This is added onto each of the materials offset
Render Queue	Changes when the material is drawn
Global Illuminattion	Controls if the global illumination is baked or realtime 0: Realtime 1: Baked 2: None
Double Sided Global Illumination	If the lightmapper accounts for both sides of the geometry when calculating Global Illumination
Specular Highlights	When enabled, the Material reflects the shine from direct lighting
Environment Reflections	When enabled, the Material samples reflections from the nearest Reflection Probes or Lighting Probe
Array Settings	Shows a dropdown with buttons to modify each of the texture arrays that the textures are stored in AVTextures: Albedo (rgb), Variation (a) NSOTextures: Normal (rg), Smoothness (b), Occlussion (a) EMTextures: Emission (rgb), Metallic (a) BMTextures: Blend Mask (r)
Ping Data Folder	Pings the data folder

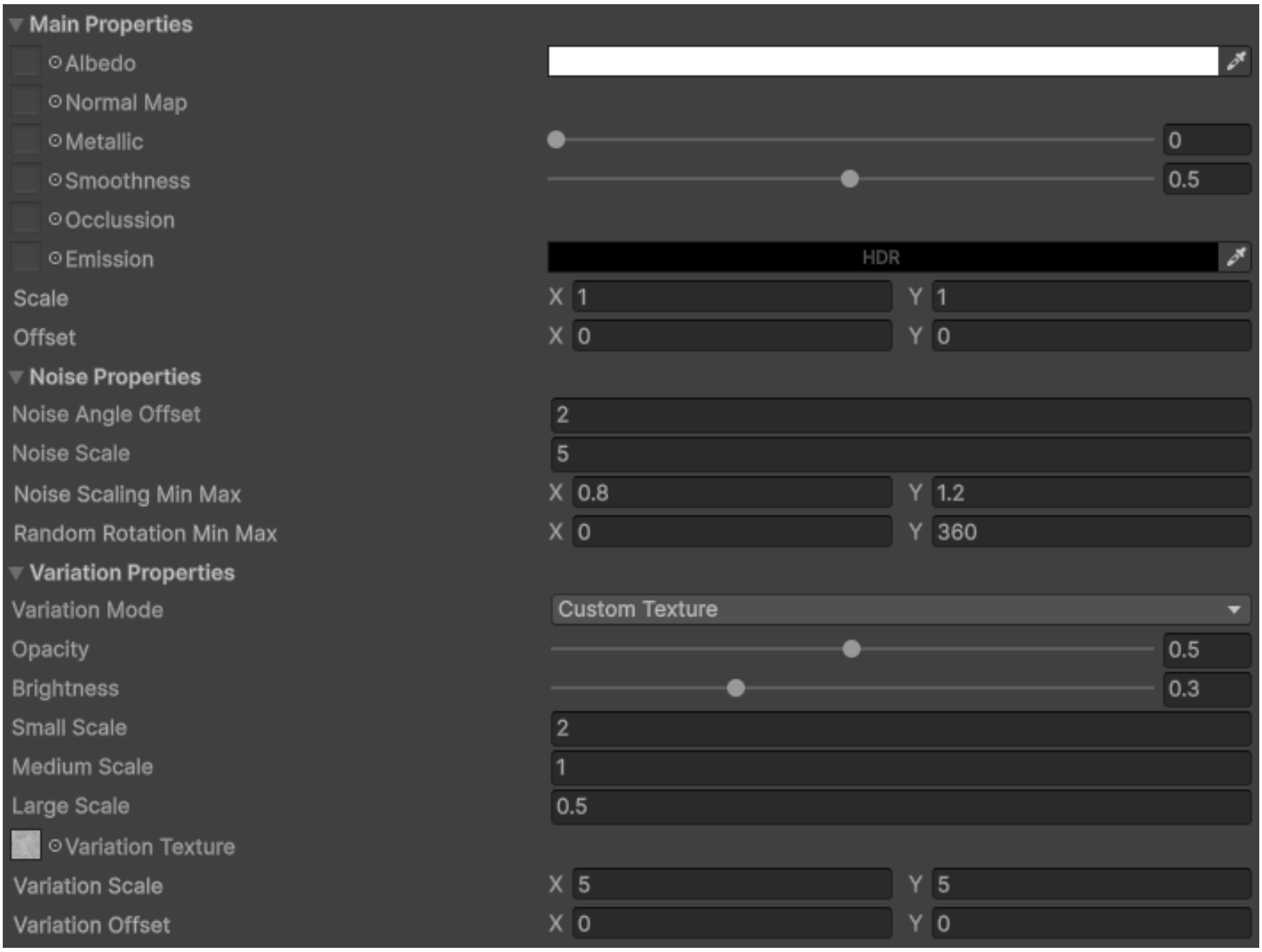
9.2 Toolbar

Each material has a toolbar with various settings

Each settings text will shrink to the first letters of each word if the inspector width is too small for the whole words to fit

Setting	Description
Noise	Adds random scaling & rotation based on voronoi noise
Random Scaling	Adds random scaling to each voronoi cell *Only shown if Noise is enabled*
Random Rotation	Adds random rotation to each voronoi cell *Only shown if Noise is enabled*
Variation	Adds random variation on top of the albedo color **Using a custom texture can cause visible tiling**
Packed Texture	If you are using a packed texture of multiple regular ones (Better for performance) R: Metallic G: Occlussion A: Smoothness/Roughness
Emission	If Emission is enabled
Smooth/Rough	Toggles between the smoothness and roughness texture **If you are using a packed texture it will sample it with a smoothness or roughness texture based on this toggle**

9.3 Material Settings



Contains all the settings for a material. Settings can be enabled and disabled through the toolbar

This is shown for each enabled material (Base material, distance blend material, blend material)

When a texture is assigned for fields that only use one texture channel, it will have a dropdown where you can choose which channel to use

9.3.1 Main Properties

Property	Description
Albedo	Albedo (RGB)
NormalMap	Normal map (RG)
Metallic	Metallic (R)
Smoothness/Roughness	Smoothness/Roughness (R) *Changes between smoothness and roughness based on toolbar setting*
Occlussion	Occlussion (R)
Emission	Emission (RGB)
Packed Texture	The mask map for the current material R: Metallic G: Occlussion A: Smoothness/Roughness *- Only shown if Packed Texture is enabled* *- Alpha toggles between smoothness and roughness based on toolbar setting*
Occlussion Strength	The occlussion strength of the packed texture occlussion *Only shown if Packed Texture is enabled*
Scale	The texture tiling
Offset	The texture offset

9.3.2 Noise Properties

These are only shown if Noise is enabled in the toolbar

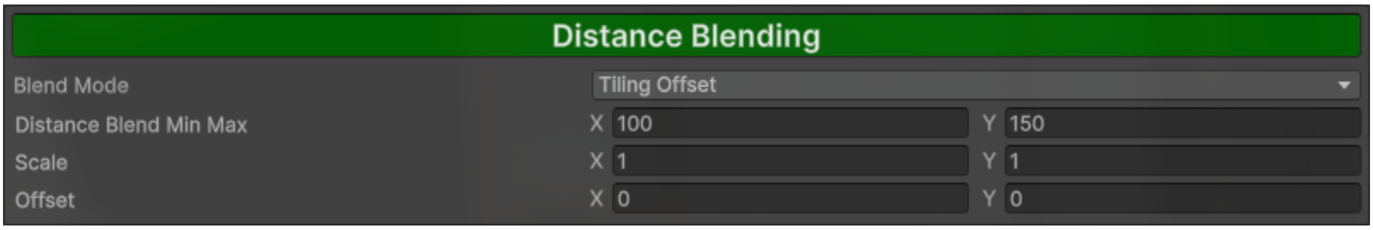
Property	Description
Noise Angle Offset	The angle offset of the voronoi noise *Only shown if noise quality is set to high*
Noise Scale	The scale of the voronoi noise
Noise Scaling Min Max	Range that each voronoi cell is randomly scaled by x: Min Scale y: Max Scale *Only shown if Random Scaling is enabled in the toolbar*
Random Rotation Min Max	Range that each voronoi cell is randomly rotated by x: Min Rotation Degrees y: Max Rotation Degrees *Only shown if Random Rotation is enabled in the toolbar*

9.3.3 Variation Properties

These are only shown if Variation is enabled in the toolbar

Property	Description
Variation Mode	The variation mode used Using a custom texture can cause visible tiling 0: Perlin noise 1: Simplex noise 2: Custom texture
Opacity	Transparency of the variation
Brightness	Intensity of the variation
Small Scale	Scale of the small variation sample
Medium Scale	Scale of the medium variation sample
Large Scale	Scale of the large variation sample
Noise Strength	Strength of the noise *Only shown if Variation Mode is set to any noise*
Noise Scale	The noise tiling *Only shown if Variation Mode is set to any noise*
Noise Offset	The noise offset *Only shown if Variation Mode is set to any noise*
Variation Texture	Texture that is drawn onto other materials, can cause visible tiling Variation (R), other channels are ignored *Only shown if Variation Mode is set to Custom Texture*
Variation Scale	The variation tiling *Only shown if Variation Mode is set to Custom Texture*
Variation Offset	The variation offset *Only shown if Variation Mode is set to Custom Texture*

9.4 Distance Blending

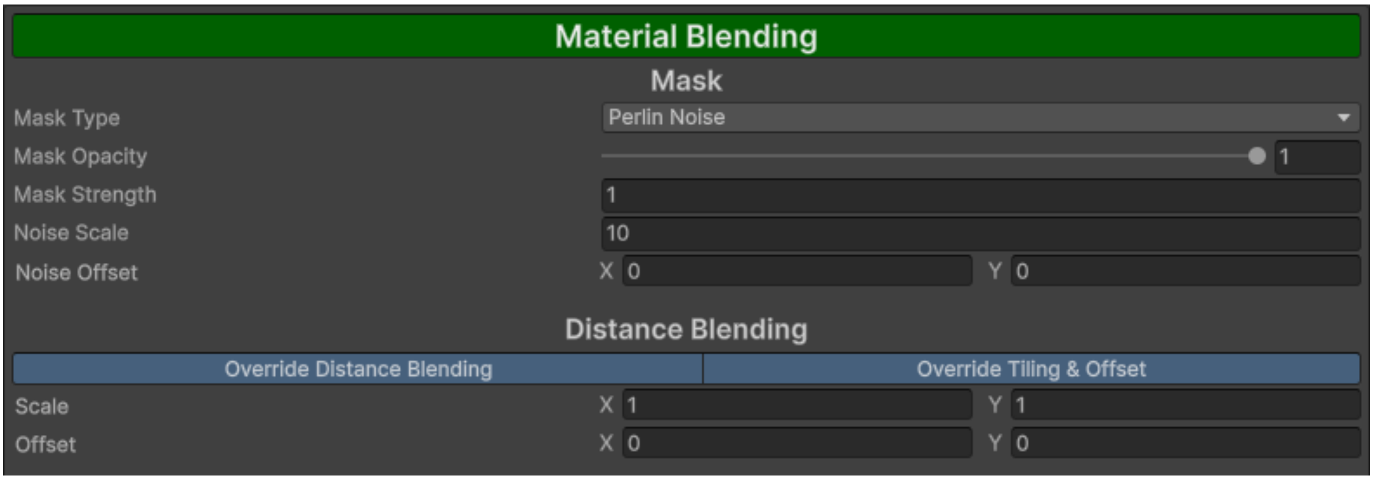


This is the material that will fade into the distance based on Distance Blend Min Max. Can be toggled by clicking the Distance Blending button

If the Blend Mode is set to Material, a separate Material Settings will be shown below for the far material

Property	Description
Blend Mode	Tiling Offset: Resamples materials with different Tiling and Offset Material: Samples a seperate far material
Distance Blend Min Max	Blend distance which the material will be sampled. Materials will be blended with regular material at min, and far material at max x: Min Distance y: Max Distance
Scale	The far tiling *Only shown if Blend mode is set to Tiling Offset*
Offset	The far offset *Only shown if Blend mode is set to Tiling Offset*

9.5 Material Blending



This is a separate material that will overlay the base and far material if set based on a mask. Can be toggled by clicking the Material Blending button

9.5.1 Mask Settings

Property	Description
Mask Type	The mask type used 0: Perlin noise 1: Simplex noise 2: Custom texture 3: Vertex Colour
Mask Opacity	Opacity of the mask and in response the blend material
Mask Strength	The higher the value, the sharper the edges and vice versa
Noise Scale	The noise tiling *Only shown if Mask Type is set to any noise*
Noise Offset	The noise offset *Only shown if Mask Type is set to any noise*
Blend Mask	Texture that is sampled as the mask for the blend material. Color from black-white represents opacity (0-1) Blend Mask (R), other channels are ignored *Only shown if Mask Type is set to Custom Texture*
Scale	The blend mask tiling *Only shown if Mask Type is set to Custom Texture*
Offset	The blend mask offset *Only shown if Mask Type is set to Custom Texture*

If mask type is set to any noise

Property	Description
Noise Scale	The noise tiling
Noise Offset	The noise offset

If mask type is set to Custom Texture

Property	Description
Blend Mask	Texture that is sampled as the mask for the blend material. Color from black-white represents opacity (0-1) Blend Mask (R), other channels are ignored
Scale	The blend mask tiling
Offset	The blend mask offset

If mask type is set to Vertex Colour

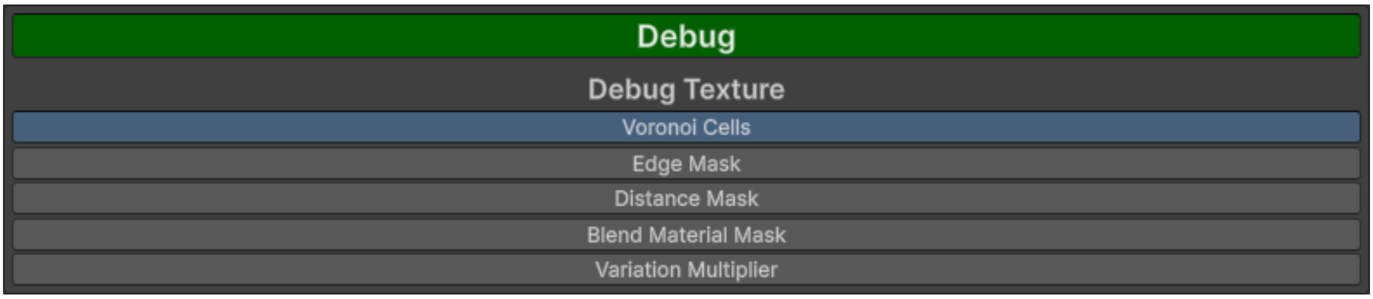
Property	Description
Mask Colour	The colour used as a mask in the vertex colour
Colour Threshold	The smoothstep min max for the colour

9.5.2 Distance Blending Settings

These are only shown if distance blending is enabled

Property	Description
Override Distance Blending	Draws the blend material on top of the far material
Override Tiling & Offset	Uses defined Tiling & Offset rather than distance blend Tiling & Offset
Scale	The blend mask distance scale *Only shown if Override Tiling & Offset is enabled*
Offset	The blend mask distance offset *Only shown if Override Tiling & Offset is enabled*

9.6 Debug



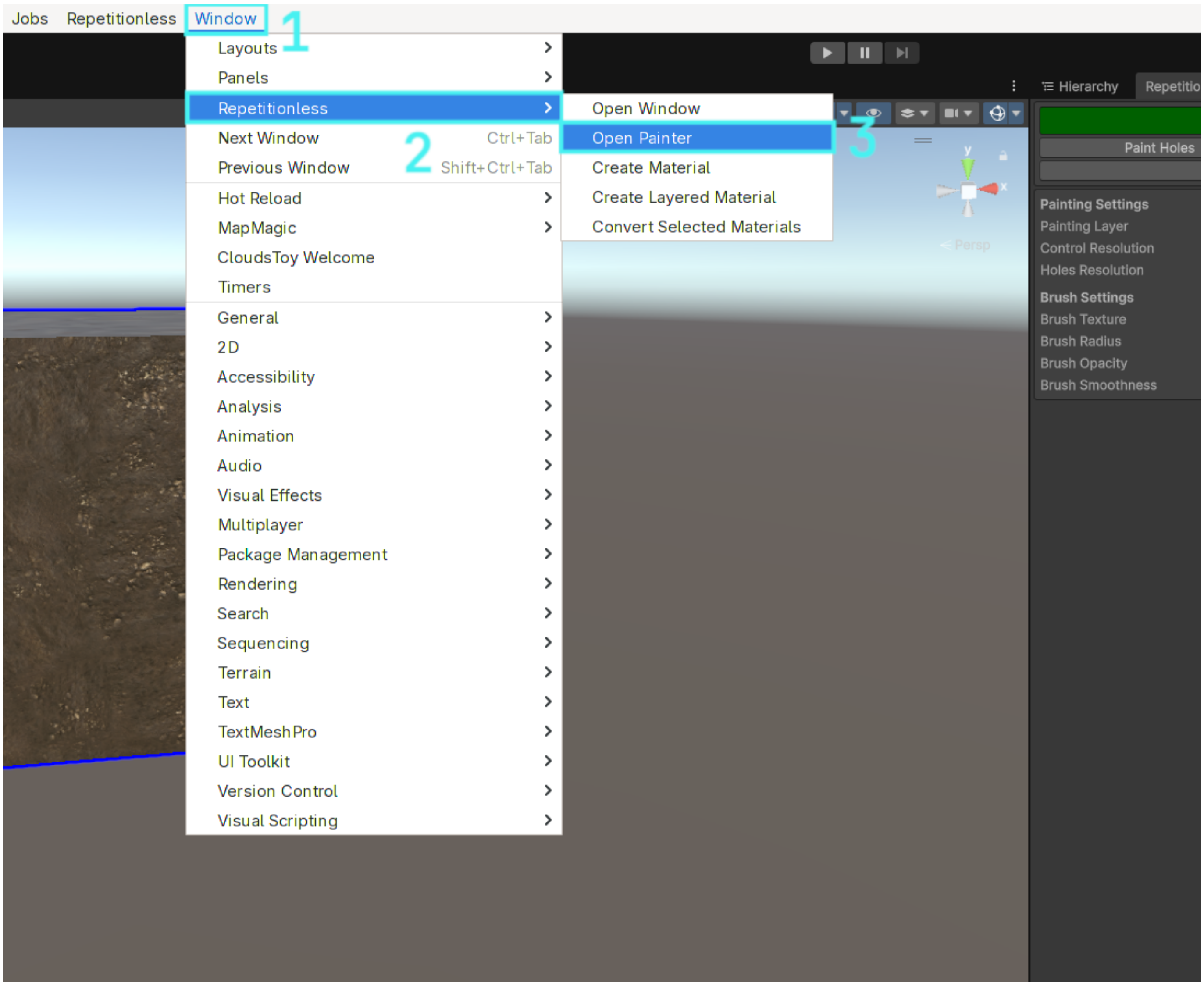
This will show the values of the selected option. Can be toggled by clicking the Debug button

Option	Description
Voronoi Cells	Outputs the value of each voronoi cell
Edge Mask	Outputs the edge mask for the voronoi cells
Distance Mask	Outputs the distance mask
Blend Material Mask	Outputs the blend material mask
Variation Multiplier	Outputs the variation multiplier

10. Painting

10.1 Setting Up

You can open the painter window in the toolbar:
`Window > Repetitionless > Open Painter`

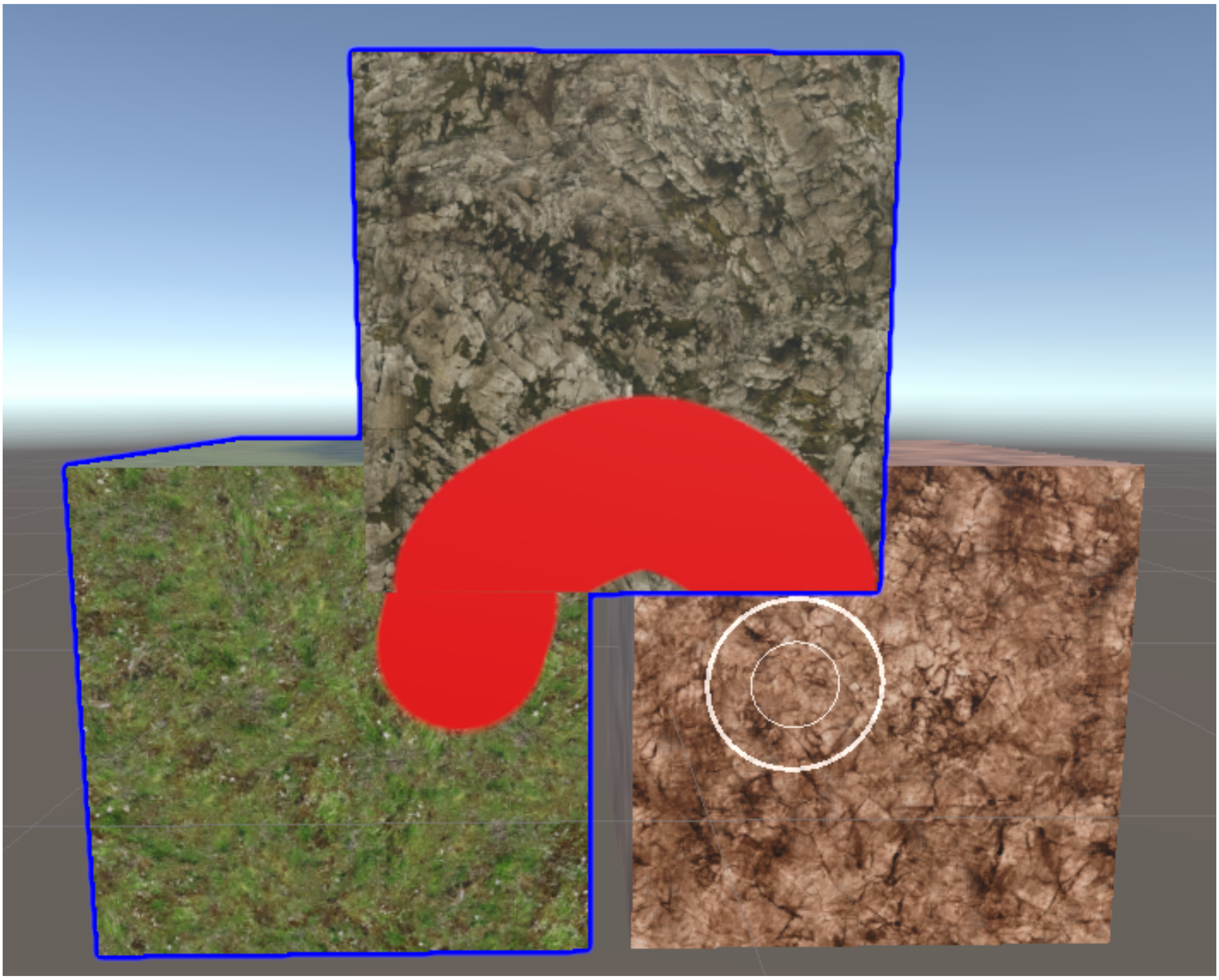


Painting is only supported on objects that have the following:

- A mesh renderer with a Repetitionless Layered material (If the layer mode is not set to control textures it will automatically be converted)
- A mesh collider

Trying to paint any other object will do nothing

10.2 Selection



When painting is enabled, clicking an object will automatically select it and start painting if the object can be painted

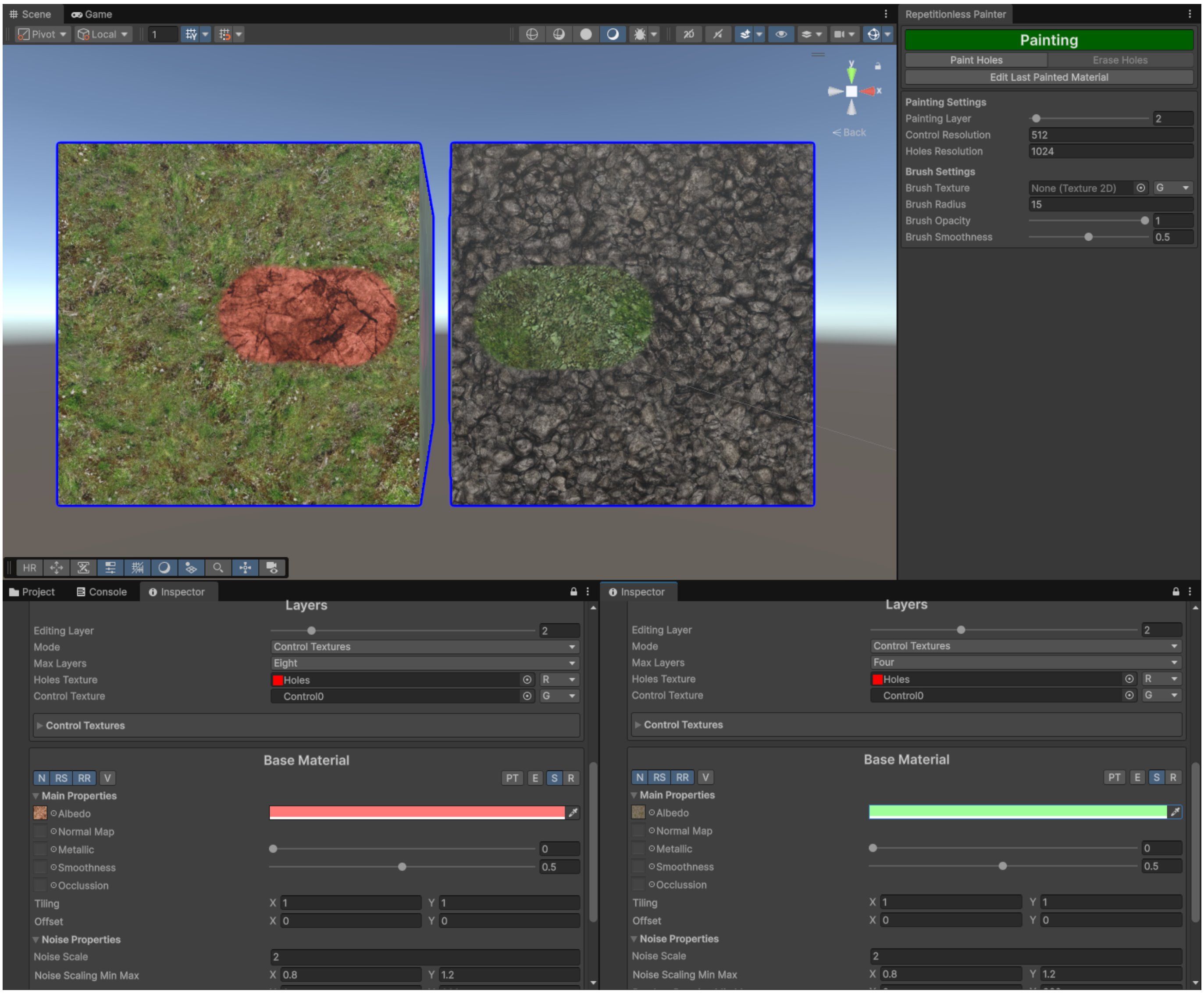
Objects that are selected for painting will have a dark blue outline around them. Trying to paint outside of any selected objects in one stroke will do nothing even if the unselected hovered object can be painted. It basically works as a mask

Regular object selection while painting is partially disabled. When selecting objects in the scene view it will use the painter selection, but you can select objects normally in the Hierarchy window

When an object is first selected, all the control textures will be created/resized, and the holes texture will be created/resized. This can take some time depending on your control/holes resolution settings

10.3 Layers

The layer selection in the window is the selected layer for each material. This is per material and can be edited in its inspector. For example, layer 1 on one object could be different to layer 1 on another object as shown below

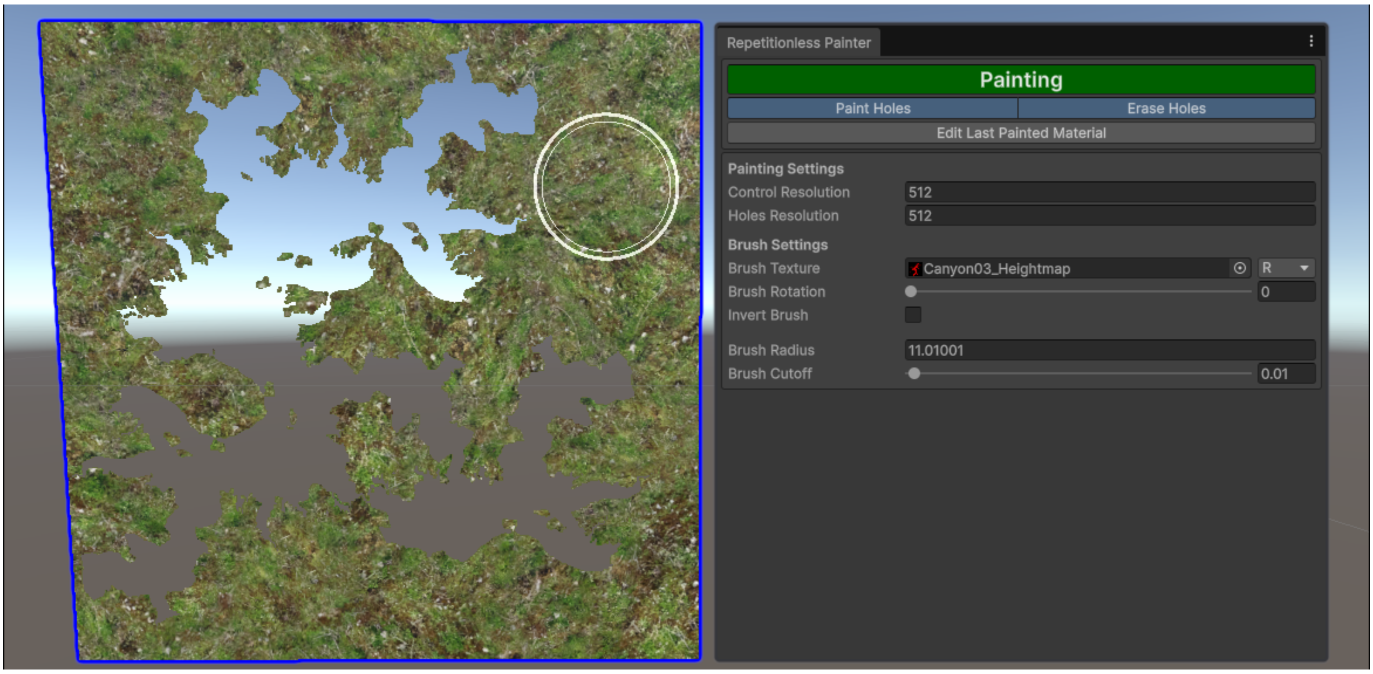


10.4 Holes

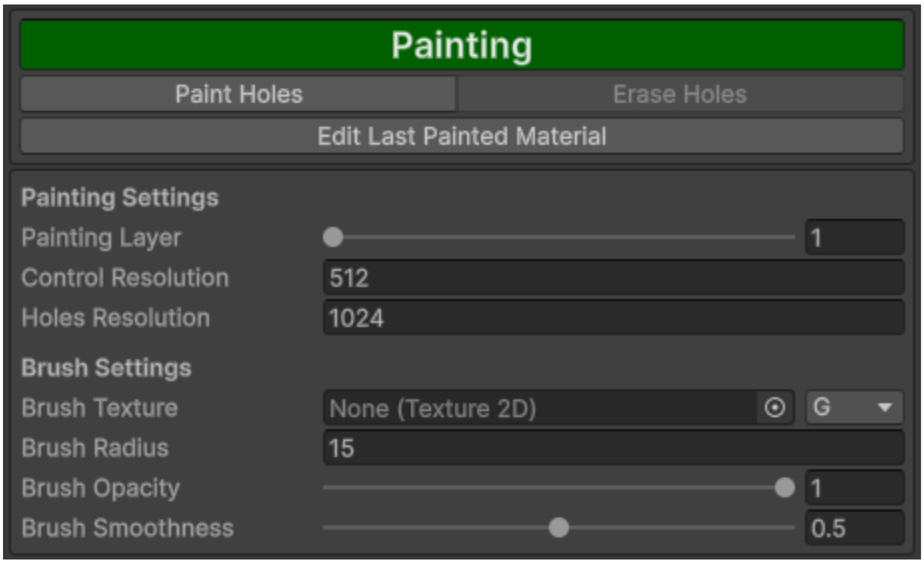
Holes can be painted onto any object with the Paint Holes toggle

When Erase holes is enabled, painting over existing holes will erase them. This is visualised with an inner circle inside the brush when enabled

Since holes do not have transparency, when using a brush texture, you can adjust the cutoff to determine which parts of the texture will be counted as a hole. If a value at a given pixel is less than the cutoff, it will not be counted as a hole


Repetitionless Painter
Painting
Paint Holes
Erase Holes
Edit Last Painted Material
Painting Settings
Control Resolution
512
Holes Resolution
512
Brush Settings
Brush Texture
Canyon03_Heightmap
R
Brush Rotation
0
Invert Brush
Brush Radius
11.01001
Brush Cutoff
0.01

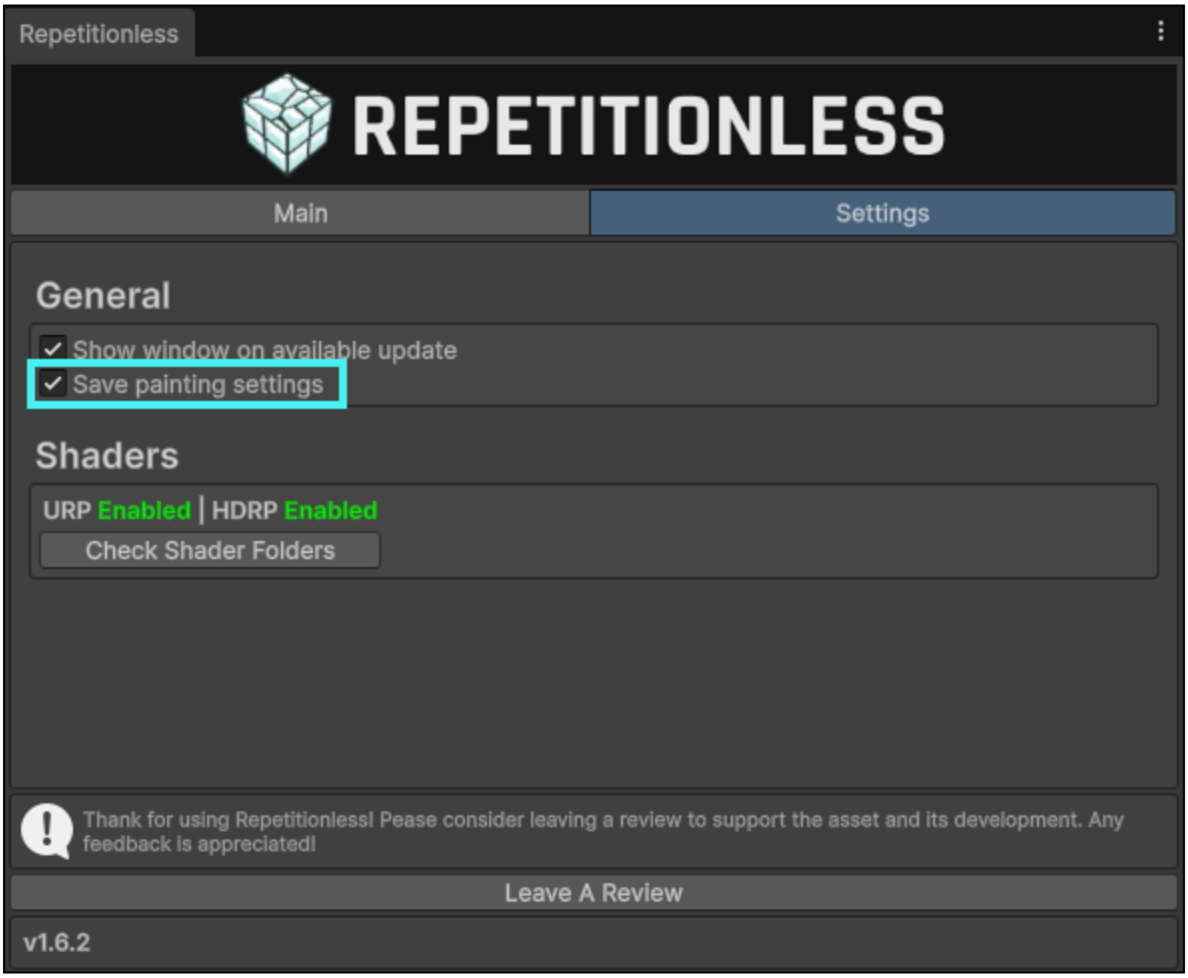
10.5 Properties



Property	Description
Painting Layer	The layer that will be painted. This is determined per material in its layer selection
Control Resolution	**Default:** 512 The resolution of the control textures. Existing textures will be automatically resized to this resolution
Holes Resolution	**Default:** 512 The resolution of the holes texture. Existing textures will be automatically resized to this resolution
Brush Texture	The texture used for painting, if not set it will be a circle filling the radius. The channel is what channel to read from in the texture
Brush Rotation	The rotation of the brush texture relative to the uvs of the painted object. This does nothing with no texture set
Invert Brush	Inverts the brush texture, sampling it as (1 - value)
Brush Radius	The size of the brush
Brush Cutoff	The cutoff of the value for the brush texture where holes will not be drawn
Brush Opacity	he strength of the brush. The brush will accumulate so if you want opacity while dragging, set this to <= 0.05
Brush Smoothness	What radius to start fading out the brush. This is visualised as the inner circle in the scene view

All properties can be auto saved and kept between painting sessions (enabled by default). You can toggle this by:

1. Opening the main window in the toolbar at `Window > Repetitionless > Open`
2. Click the settings tab
3. Toggle the setting called `Save painting settings`



10.6 Keybinds

When using keybinds, they will always show with a popup next to the mouse so you know what you have just changed without looking at the window. This allows painting objects without having the window visible at all

Note that:

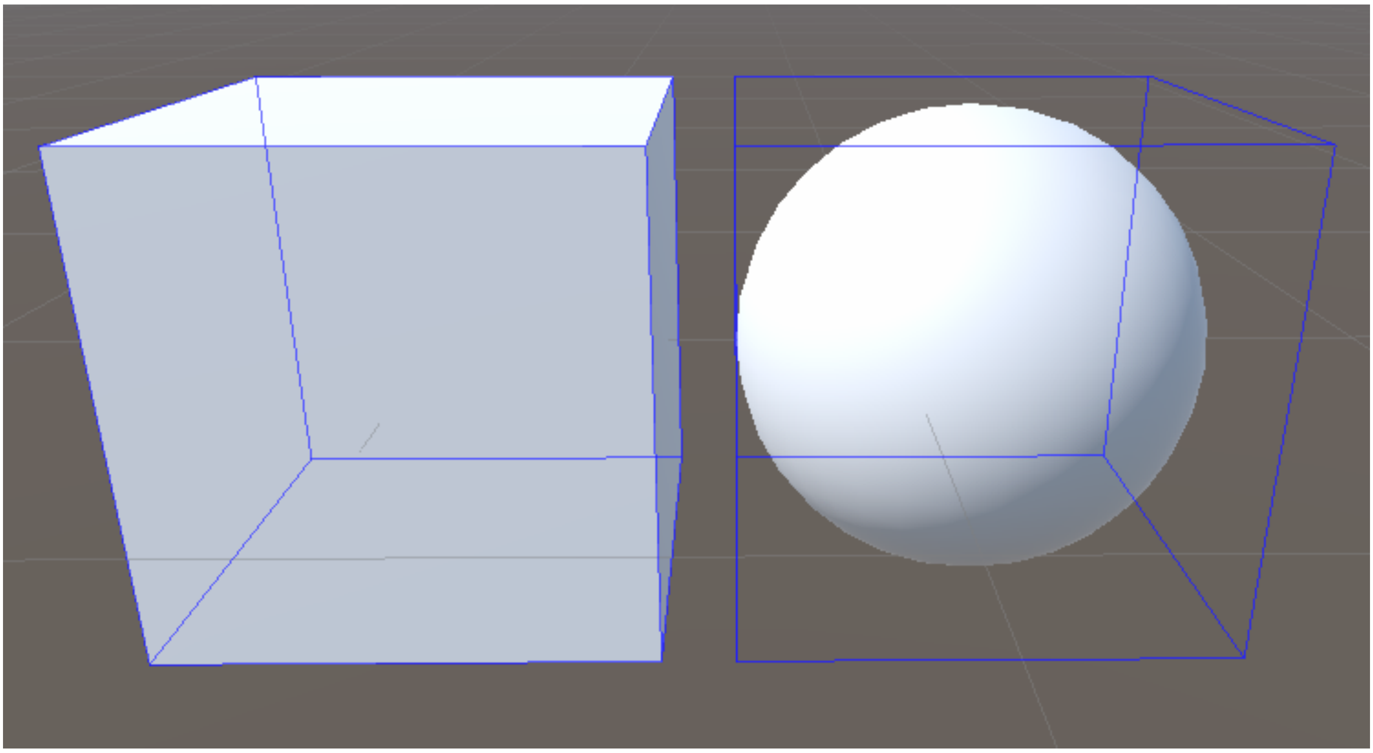
- Keybinds will only work when the scene view is focused
- Default actions with the below keybinds will be disabled while painting.

Key	Action
A	**Painting Control:** Changes the brush opacity **Painting Holes:** Changes the brush cutoff
S	Changes the brush Radius
D	**Painting Control:** Changes the brush smoothness **Painting Holes**: Toggles erasing holes
C	Changes the brush texture rotation
F	Focuses the scene camera to the mouse position
G	Toggles painting
H	Toggles painting holes
Shift + Resize	**Compatible with Opacity, Cutoff, Radius, Smoothness, & Rotation controls** Slows down the modification of a slideable property to x0.1 the default. Also shows an extra digit in the mouse popup
Shift + Click	Deselects the hovered object
Shift + Scroll	Changes the painting layer
Control (Hold)	Inverts the brush

10.7 Caveats

10.7.1 Pre Unity 2022

Below Unity 2022 there is no cheap built in way to draw outlines around the selected objects. In these versions a wire cube is drawn around the bounds of selected objects instead.

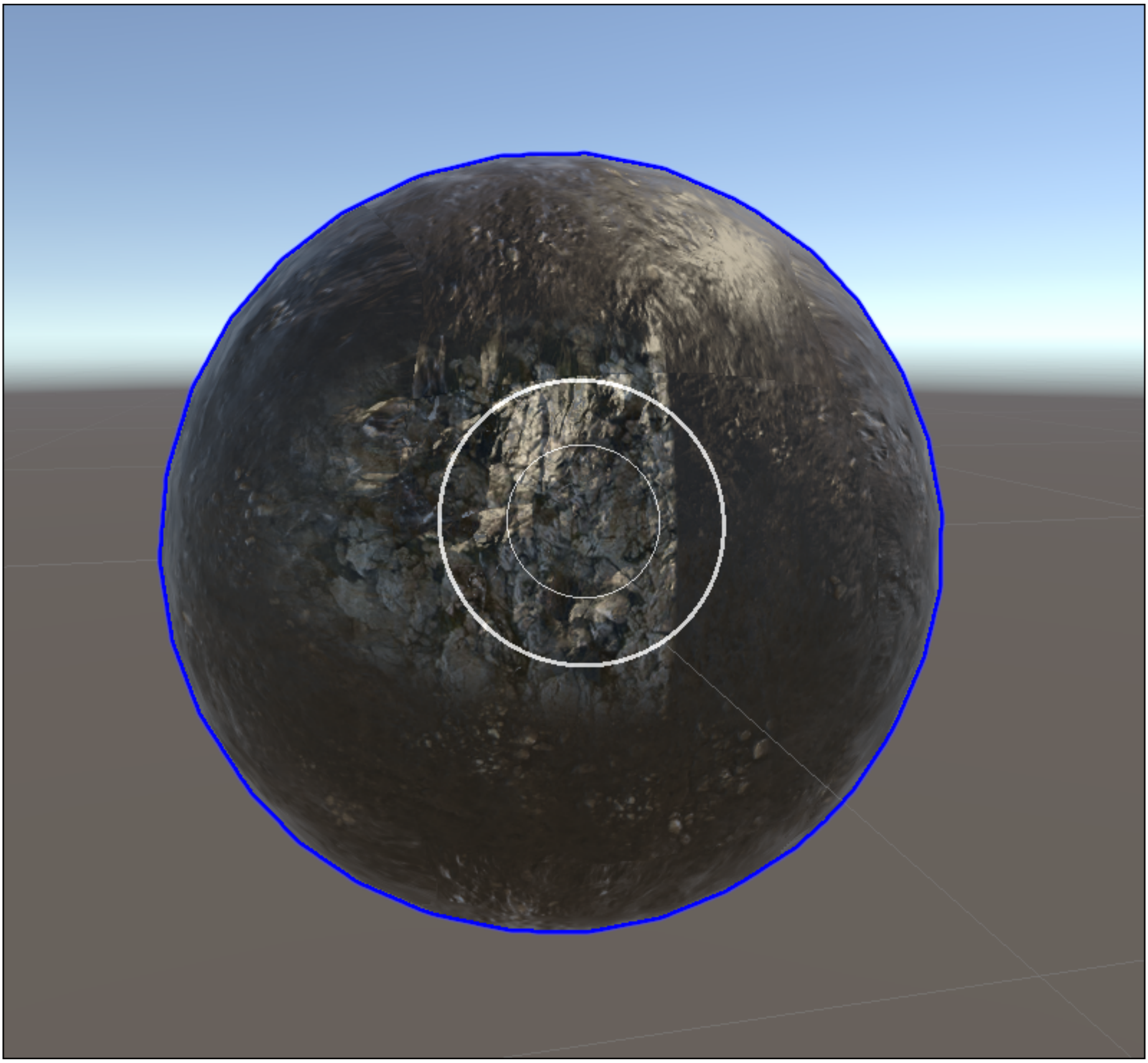


10.7.2 UVs

The UVs on a mesh should have uniform texel density otherwise painting may be distorted and not fit within the brush preview. Basically, the better unwrapped the UVs, the better the painting will be

The painter will not paint across different UV tiles. If you paint on one and the radius overlaps another, it will only paint on the hovered tile and show a visible seam

Below is an example of the default unity sphere showing the above issues:

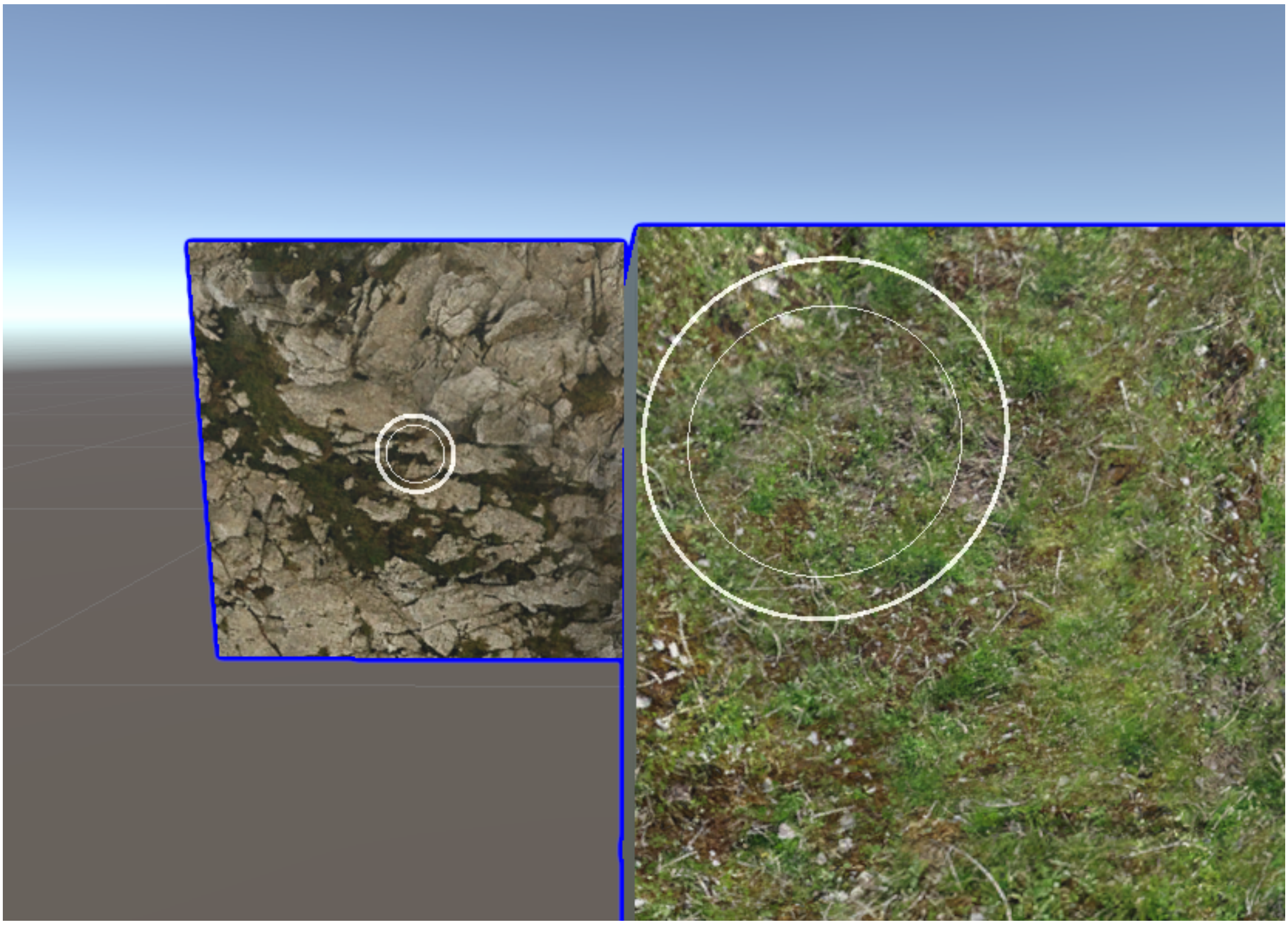


10.7.3 Scaling

Objects with a non-uniform scale will have their paint stretched, similar to the above UV stretching issue.

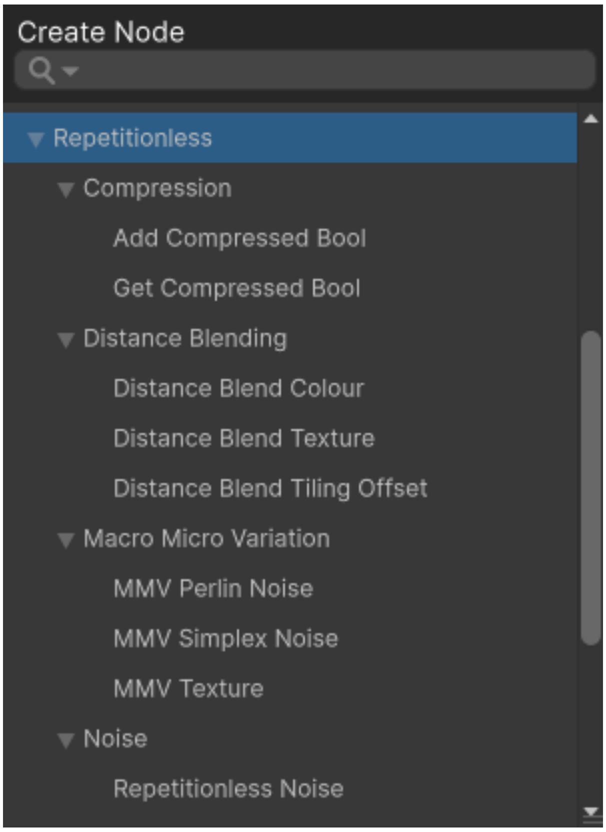
When an object is smaller or larger than 1x1x1, the brush radius will correspond to that size as the UVs will change with it. To combat this the brush preview radius will change with the object but if the scale is non-uniform it may show incorrect.

Below is an example with the same radius on two different scaled objects



11. Shader Creation

11.1 Shader Graphs



To create a shader graph using the included sub-shaders, you can find the sub shaders in the shader graph under `Create Node > Repetitionless`

To view details about each sub-shader, view the corresponding Sub Graphs page in the contents on the left of the page

11.2 Shader Code

For creating your own shaders using the included shaders, view the respective Shader API pages in the contents on the left of the page for details on each set of functions and how to implement them into your own code

Shaders can be found at `Packages/com.williamschack.repetitionless/Shaders/HLSL`

12. Scripting

For creating your own scripts and inspectors using the included scripts, view the respective Scripting API pages in the contents on the left of the page for details on each set of functions and how to implement them into your own code

Scripts can be found at `Packages/com.williamschack.repetitionless` in the `Editor` and `Runtime` folders

To include script namespaces you must reference their respective assembly definitions in your own assembly definitions

13. Submitting Issues

If you would like to submit a Bug Report, Feature Request, or Question, you can:

- Submit an issue to the discord (https://discord.gg/2NTvBDB3xq))
- Submit an issue to the github repo (github.com/WilliamSchack/Repetitionless-Issues)
- Email me at (support@wilschack.dev)

14. Integrations

14.1 MapMagic

`MapMagic2 Must Be Installed`

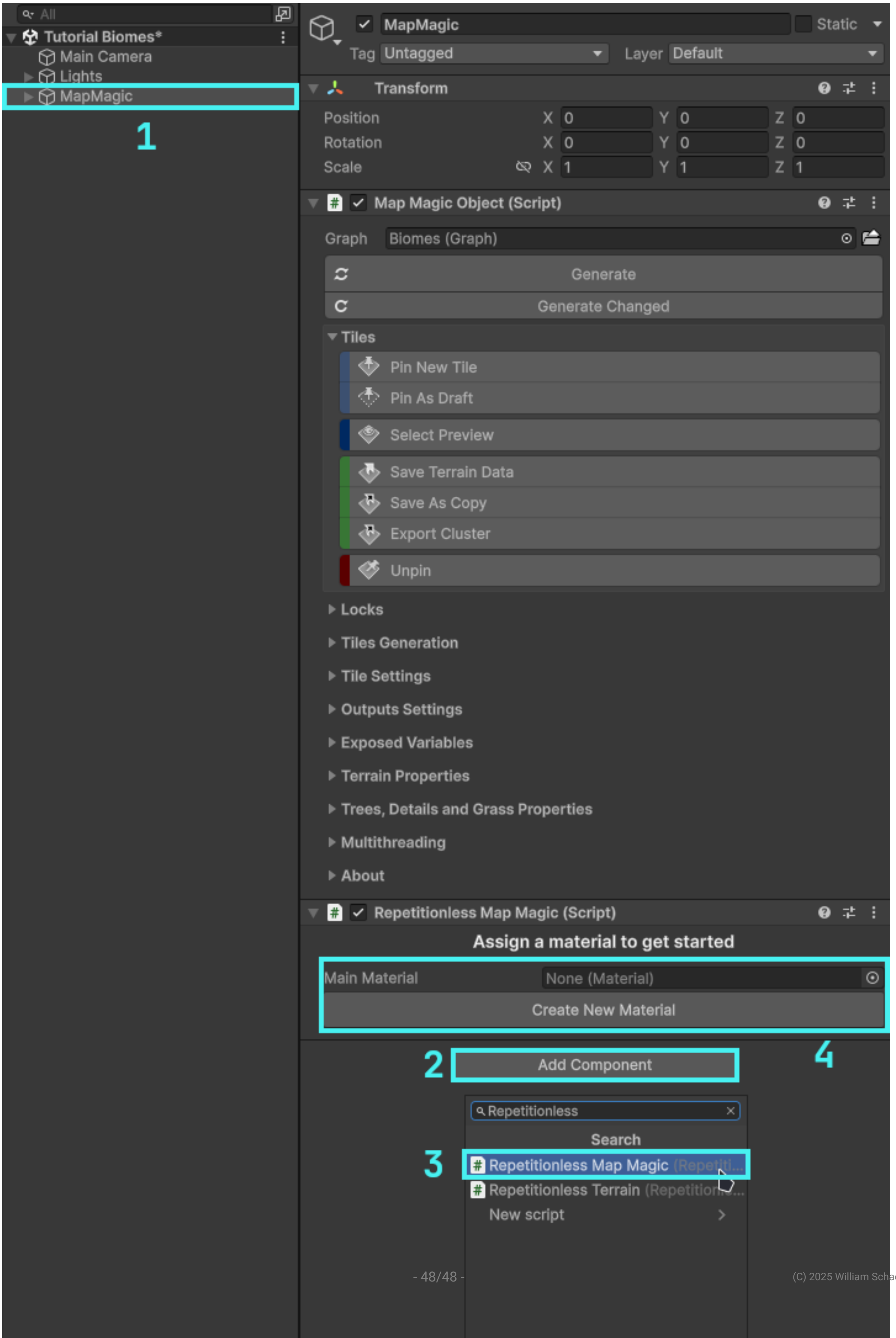
14.1.1 Applying Repetitionless

To use repetitionless on a MapMagic terrain, add a `RepetitionlessMapMagic` component to the object that has the MapMagicObject

1. Select the map magic terrain you want to use
2. Select `Add Component`
3. Select `Repetitionless Map Magic`

To add a material you can either:

- Click the `Create New Material` button
- Assign a material in the `Main Material` field


All
Tutorial Biomes*
Main Camera
Lights
MapMagic
1
MapMagic
Static
Tag Untagged
Layer Default
Transform
Position X 0 Y 0 Z 0
Rotation X 0 Y 0 Z 0
Scale X 1 Y 1 Z 1
Map Magic Object (Script)
Graph Biomes (Graph)
Generate
Generate Changed
Tiles
Pin New Tile
Pin As Draft
Select Preview
Save Terrain Data
Save As Copy
Export Cluster
Unpin
Locks
Tiles Generation
Tile Settings
Outputs Settings
Exposed Variables
Terrain Properties
Trees, Details and Grass Properties
Multithreading
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New script